FINITE CERTIFICATES AND UNDECIDABLE GIBBS UNIQUENESS

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Abstract. We construct translation-invariant classical interactions on $\mathbb{Z}^2$ for which deciding uniqueness among all Gibbs states is undecidable at inverse temperature one. The finite on-site alphabet depends effectively on the input; interaction supports are fixed, and local energy terms take only the integer values $0, 1, 20$. Exact character orthogonality converts a strictly positive finite-energy model into a gas of finite computation histories. A halting certificate produces a uniformly positive phase-selection bias, however small. A reference Ising comparison and an independent-disagreement argument then give uniqueness and exponential clustering of every local physical observable. Without halting, the entire Gibbs simplex is affinely isomorphic to the two-dimensional zero-field Ising simplex. A distinct soft-certificate construction has the same dichotomy for one table at every finite inverse temperature at least $2/5$. We prove boundary-specific exponential convergence, smooth pressure with summable cumulants on the uniqueness branch, uniform computability of the selected phase and its right pressure jet, and stability under small finite-range spin perturbations. An effective compactness argument supplies a computable valid convergence modulus for each unique computable finite-range specification; the impossibility concerns total bounds across the undecidable family. Exact microscopic symmetrization excludes a family-wide perturbation radius for arbitrary real interactions.

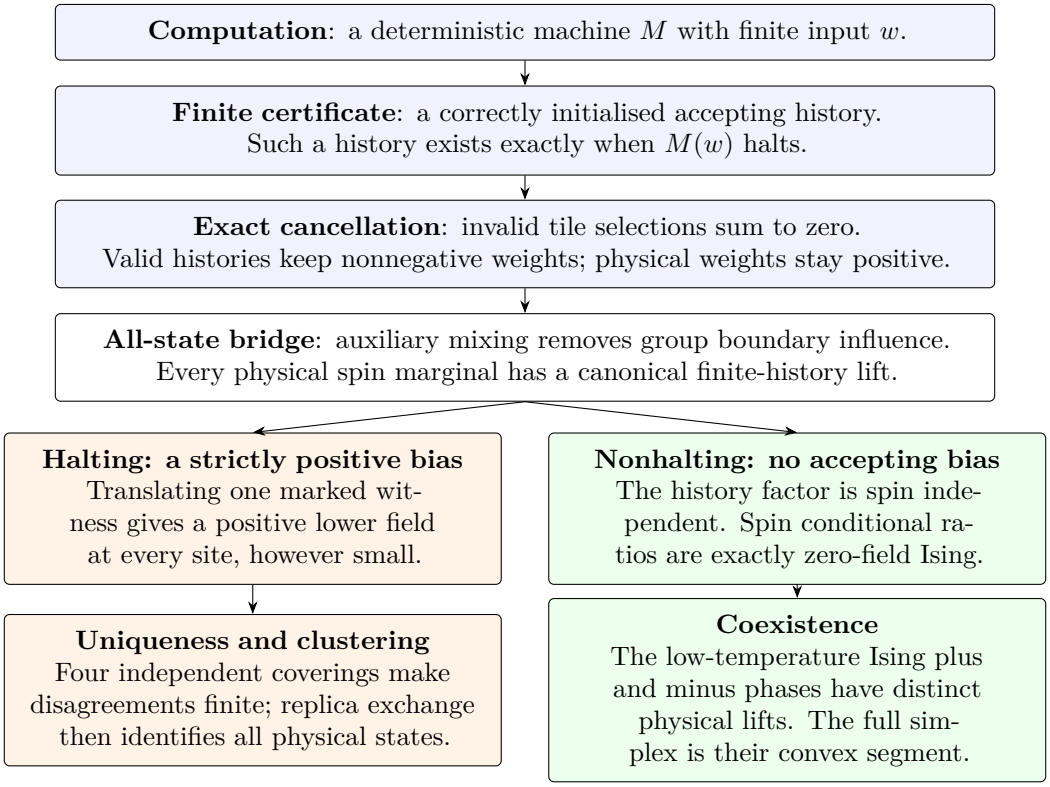
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Conceptual overview

The main construction turns one question about a computation into one question about equilibrium: does the entire physical Gibbs simplex contain one state or more than one? The compiler outputs a finite interaction without deciding which outcome occurs. At inverse temperature $\beta = 1$, its physical weights are strictly positive, its local energy terms lie in $\{0, 1, 20\}$, and only the finite on-site alphabet depends on the machine and input.



Where exactness matters. An arbitrarily small uncompensated accepting weight would already produce a false positive field on a nonhalting input. Example 2.1 computes this error explicitly. Cancellation concerns signed terms in an expansion of positive weights; it does not forbid any physical configuration.

Construction, refinements and established inputs. The exact-table reduction is the principal construction (theorem 1.1). A separate soft table and its common temperature interval refine it (theorem 1.2); convergence, pressure and perturbation results come afterwards (theorem 1.3). Ising uniqueness and classification, Dobrushin comparison and finite replica exchange are established machinery. The contribution is their verified connection to the finite-temperature computational encoding and to *all* physical Gibbs states.

1. Introduction and principal results

Uniqueness of a Gibbs state asks whether every boundary condition leads to the same infinite-volume equilibrium. It is a statement about an entire specification, not merely about decay of correlations in one chosen phase. At low temperature the two-dimensional Ising model illustrates the distinction: its plus phase clusters, yet a different minus phase exists. We show that this distinction can encode an arbitrary halting problem even when the microscopic data are exact integers and the temperature is fixed in advance.

The principal construction is theorem 1.1. Its contribution is the conjunction of fixed positive temperature, strictly positive physical weights, a fixed finite integer-energy palette, and a halting reduction for uniqueness among *all* DLR states. Computation in lattice rules and the probabilistic tools used below have earlier precedents; the finite-table encoding and its all-state physical projection are the interfaces proved here. The conceptual overview displays the complete argument before its technical development.

The computational information enters through finite certificates: correctly initialised finite histories containing a halting event. Directly rewarding a local approximation to such a history is inadequate at positive temperature, where every finite pattern, including an erroneous one, has positive probability. Instead we arrange an exact algebraic cancellation. Auxiliary finite-group characters cancel every finite tile selection with an unmatched boundary or inconsistent local colours. The surviving terms are nonnegative weights of valid finite computation histories. The microscopic model nevertheless has no forbidden configurations and no infinite energy.

A certificate may be extremely large, and its induced bias arbitrarily small. A uniform high-temperature estimate for the distinguished spins is consequently unavailable. We use the bias only to select the plus phase of a low-temperature Ising reference. The reference supplies rare contours. Independently, computation histories are covered by a dilute family of finite objects. Two replicas yield four independent covering families. Their subcritical overlap graph makes every disagreement component finite, permitting a finite replica exchange and proving uniqueness for arbitrary Gibbs states. The exchange principle is established machinery; the independent covering construction and its connection to the positive physical model are additional steps. Example 2.1 isolates the need for an exact zero, and section 4.1 explains the geometry of the coverings and replica exchange before the estimates.

1.1. **Conventions.**

An interaction on a finite alphabet $\mathcal{A}$ assigns finite real functions Φ_X to finite $X \subset \mathbb{Z}^2$. Our interactions are translation invariant and have fixed finite range. For finite Λ, H_Λ^τ is the sum of terms meeting Λ, with exterior configuration τ. The Gibbs kernel is proportional to $\exp(-\beta H_\Lambda^\tau)$; a Gibbs state satisfies all these local conditional identities (the DLR equations). No invariance is imposed on a Gibbs state. A dimensionless field multiplies the weight by $\exp(\sum_x H_x s_x)$.

Distances and diameters use the ℓ^∞ metric, unless described as nearest-neighbour adjacency. For finite A, A^+ is its closed nearest-neighbour collar: $|A^+| \le 5|A|$ and $\operatorname{diam} A^+ \le \operatorname{diam} A + 2$. Put $K = 20\beta$. An Ising bond contributes $\exp(-K\mathbf{1}_{s_x \neq s_y})$; the conventional exponent coupling is $J = K/2$.

Effectiveness means that a finite table is computable from the input, not that its construction is efficient. Computability assertions require computable parameters, or effective names for them. Qualitative assertions hold for arbitrary real parameters in the stated ranges. Computability of a state means computability of its cylinder probabilities, equivalently of expectations of local observables with computable finite tables; covariance bounds allow arbitrary bounded local observables.

1.2. **Main theorem: the integer-energy construction.**

The first theorem contains the undecidability construction. Its input is a finite machine description, and its output is an exact finite physical interaction; no coupling is specified by an oracle for a halting time.

Theorem 1.1 (Fixed-temperature undecidability)**.** *An algorithm takes a deterministic Turing machine and its finite input to a finite alphabet with a distinguished coordinate* $s_x \in \{-1, 1\}$ *and a translation-invariant interaction on* $\mathbb{Z}^2$ *with support shapes*

$$\{0, -e_1, -e_2\}, \qquad \{0, e_1\}, \qquad \{0, e_2\}.$$

The displayed local terms have energy values in $\{0, 1, 20\}$*. At* $\beta = 1$*:*

(i) *Without halting, spin projection is an affine bijection from the entire physical Gibbs simplex onto the zero-field nearest-neighbour Ising Gibbs simplex with mismatch exponent* 20*.*

(ii) *With halting, there is exactly one physical Gibbs state. Every pair of bounded local physical observables has exponentially decaying connected correlation.*

Thus full Gibbs uniqueness is undecidable for these exact finite tables. Without halting there are exactly two extremal Gibbs states; their equal mixture has spin covariance bounded away from zero at arbitrary separation.

1.3. **A refinement: soft certificates and a temperature interval.** Each accepting history in theorem 1.1 imposes a hard plus pin at a single anchor. This constraint belongs to a proof representation, not to the physical model. The next result refines the construction by using a *different* table: a finite penalty replaces each proof pin, unmarked histories are permitted, and one history may contain several marks. Uniform estimates then give a common temperature interval for that table. This is a further construction sharing the same mechanism, not a change to the table in theorem 1.1.

Theorem 1.2 (Soft certificates and a common temperature interval)**.** *An effective construction with the same support shapes and energy values has the following properties for the* same table *at every finite* $\beta \geq 2/5$*.*

(i) *Without halting, its entire physical Gibbs simplex is the affine lift of the zero-field Ising simplex with mismatch exponent* $K = 20\beta$*.*

(ii) *With halting, its entire physical Gibbs simplex is a singleton, exponentially clustering for all bounded local physical observables. For spin observables on* A, B*,*

$$|\operatorname{Cov}(f, g)| \leq 4\|f\|_\infty \|g\|_\infty |A||B| 2^{-\operatorname{dist}(A,B)}. \tag{1.1}$$

(iii) *At each fixed* β*, a witnessed marked history determines* $h_* = h_*(\beta, \gamma_*) > 0$ *such that uniqueness and the same spin bound hold for every bounded inhomogeneous field with* $\inf_x H_x > -h_*$*. For uniform* $h > -h_*$*, the physical pressure is* C^∞*, with absolutely convergent cumulant formulas and* $0 \leq p''_\beta(h) \leq 68$*.*

The physical clustering exponent is uniformly positive, independently of the halting time. The field margin h_* *need not be bounded below across inputs.*

At zero field, for both constructions the nonhalting simplex is a segment and the halting simplex a point. Every Gibbs state is translation invariant. Each nonhalting endpoint clusters; an interior mixture does not. The existence of a clustered phase alone therefore does not distinguish the cases.

The Ising segment classification in this statement is an application of Aizenman–Higuchi. What must be supplied for the physical model is the inverse affine projection and lift, including arbitrary, potentially non-translation-invariant Gibbs states; this is proved in section 3.3.

1.4. **Further results and corollaries.** The following results describe what can be deduced or further proved once the construction is available. They do not constitute additional encodings of computation. In particular, computable pressure and smooth-response techniques have established predecessors; the new obligations here are their precise applicability and effectiveness for this family. The perturbation result requires an additional activation argument and retains its spin-only scope.

Theorem 1.3 (Effectiveness, pressure and stability)**.** *For the models and temperatures in theorems 1.1 and 1.2:*

(i) *Plus-spin boxes with freely summed incident group variables converge exponentially to a selected physical Gibbs state, uniformly over machine inputs. This state is uniformly computable. On a halting input, convergence is also exponential for arbitrary spin boundaries with these free group variables, with model-dependent constants.*

(ii) *The pressure is uniformly computable. All right derivatives at zero exist and form a uniformly computable smooth right jet. Zero-field differentiability is equivalent to halting. The left derivative is not uniformly computable across the family.*

(iii) *On a halting input, sufficiently small finite-range spin-only multibody perturbations preserve uniqueness and exponential clustering. An explicit sufficient radius depends on a witnessed history and the perturbation range. At $\beta = 1$ there is no positive family-wide radius for arbitrary real perturbations of the physical interaction.*

Every unique computable finite-alphabet, finite-range, finite-energy specification has a computable valid all-boundary convergence modulus, found by one partial algorithm under the uniqueness promise. No total algorithm on the whole machine family can supply a valid minus-boundary or all-boundary bound on every halting input.

The boundary qualifications are part of these conclusions. We prove neither strong spatial mixing for arbitrary microscopic boundaries nor a Glauber mixing-time bound. Smooth pressure does not here imply analyticity.

1.5. **Relation to earlier work.** Computation in local lattice rules begins with the domino problem [2]. Wolfram [24] discussed undecidability and intractability in physical lattice models; Kanter [14] treated computational unpredictability in classical systems. Ground-state encodings by Gu et al. [11] and Gu–Perales [10], and the characterization of multidimensional shift entropies by Hochman–Meyerovitch [12], show the strength of computational encodings at zero temperature or with forbidden patterns. Quantum spectral-gap undecidability [5] concerns a distinct problem. None of these statements is used here as a finite-temperature uniqueness theorem. We make no general priority claim over all formulations of undecidability in statistical mechanics.

The finite-temperature construction combines a positive exact integer table, character cancellation, an all-state projection and lift, and phase selection by a finite certificate of unbounded possible size. The full Ising classification is an application of Aizenman–Higuchi in the form of Coquille–Velenik [4, Corollary 1.1]. Positive-field reference uniqueness uses Lebowitz–Martin-Löf [18, Theorem 1 and p. 278] and Lee–Yang [19]. These results are applied to the ferromagnetic reference, never to the projected target.

The replica exchange is van den Berg's independent-disagreement method [22, Theorem 1]. Disagreement percolation [23] and Dobrushin comparison [6, 21] supply its context. We give the exchange and its independent covering construction explicitly. A finite-gas exchange also constructs the canonical history gas without a cluster expansion. Kotecký–Preiss [15] and Ferrari–Fernández–Garcia loss networks [8] are applicable alternatives, not new principles.

Clustering-to-smoothness arguments are classical [17, 16]. We establish the inhomogeneous uniformity and finite-tilt limit required in this model. Computability of pressure and equilibrium states has a substantial existing theory [9, 3, 7]. Our compactness certificate concerns all DLR boundaries and identifies the convergence obstruction compatible with uniqueness. Exact applicability statements appear in appendix C.

Three representations remain distinct throughout. The physical alphabet has positive weights. Conditional on spins its group variables are uniformly mixing. Free group summation then represents spin ratios by a separate positive history gas. Only its canonical finite-component sector is used. Infinite formal tilings are not identified with extra physical Gibbs states.

2. Finite histories and positive integer energies

2.1. **Two finite-certificate grammars.** A deterministic machine can be made total with absorbing states HALT and REJECT. Fold its two-sided tape onto the nonnegative integers, storing the original cells j and $-j-1$ together. A permanent origin bit permits a radius-one rule to distinguish a track change at the origin from a move elsewhere. A proposed move through a wall becomes REJECT, even if the proposed transition would halt. This prevents finite-width artefacts from becoming certificates. The full finite grammar appears in appendix A.

Both grammars have finite labelled Wang-tile sets $\mathcal{T}$ with a zero boundary colour. A finite nonzero-edge component with zero exterior must be a solid rectangle of width and height at least two. Its bottom row is the entire prescribed input followed by blanks, and subsequent rows are its exact deterministic bounded-width evolution. A finite automaton accepting precisely Pz^* implements the bottom condition, where P is the prescribed initial word and z an ordinary blank cell.

In the *hard, single-anchor grammar*, the top row must contain exactly one halting head and no other head. The top-left tile is marked. Every finite component has one mark, and a finite component exists exactly when the computation halts. In the *soft grammar*, the top automaton accepts every word; a tile is marked precisely when its centre cell carries a halting head. Unmarked prefixes and rejecting histories are allowed. A marked component exists exactly when the computation halts. Absorption of the halting state permits several marks in one component.

Colours force the full rectangle and computation, not merely plausible transitions. Nonzero matching cannot terminate at a hole or produce a ragged boundary. Components may touch across zero-coloured edges without merging.

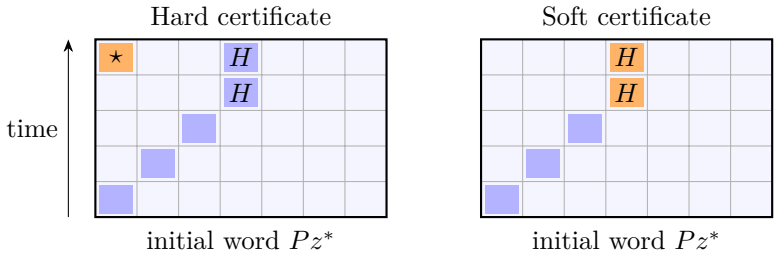


Figure 1. Two certificate tables for the same schematic history. Blue cells trace the head, and H denotes an absorbing halting head. Orange marks are the single top-left anchor in the hard table and all halting centres in the soft table. The exact finite rule is given in appendix A.

2.2. **Why a rare false certificate is still an error.**

Example 2.1 (A false accepting mark and its exact cancellation)**.** Consider a toy marker at one spin, with relative weight $w > 0$ when the spin is plus and rw when it is minus, where $0 < r < 1$. The empty marker has weight one. Summing the marker gives

$$Z_+ = 1 + w, \qquad Z_- = 1 + rw, \qquad h_{\mathrm{err}} = \frac{1}{2} \log \frac{1+w}{1+rw} > 0.$$

With $w = 1/100$ and $r = 1/2$, for instance, the false field is exactly $\frac{1}{2}\log(202/201)$. If the marker represents an invalid computation that has merely been assigned a finite penalty E, then $w = e^{-\beta E}$ remains positive at every finite E. Putting this same independent marker at every site of low-temperature zero-field Ising produces a positive uniform field after summation, and therefore selects a unique reference equilibrium. Its rarity has not preserved the intended nonhalting coexistence branch. This calculation concerns an uncompensated accepting weight; it does not rule out other encodings that correct or balance errors.

Here is the local operation that instead gives zero. Suppose neighbouring pieces of a proposed history assign different bits, 1 and 0, to their common edge. On $G = \mathbb{Z}/2\mathbb{Z}$, set $\chi_0(g) = 1$ and $\chi_1(g) = (-1)^g$. The shared-edge average distinguishes the invalid pair from a matching pair:

selected colours	$g = 0$	$g = 1$	average
$(1, 0)$	1	-1	0
$(1, 1)$	1	1	1

Thus the contradictory local claim has zero contribution, rather than a small positive one. There is no negative physical probability: for $0 < \varepsilon < 1$, both factors $1 + \varepsilon\chi_c(g)$ and $1 + \varepsilon\chi_d(g)$ are positive, while their selected cross-term $\varepsilon^2\chi_c(g)\chi_d(g)$ has zero average when $c \neq d$.

In the full construction the grammar checks initialisation, transitions and acceptance. Shared-edge cancellation enforces the required colour equalities, and dangling-edge cancellation enforces the exterior boundary. Every surviving marked object is therefore a genuine certificate. Valid unmarked histories in the soft version may survive; an invalid accepting history may not. All physical configurations still have positive weight.

2.3. **The positive physical table.** Let $t = |\mathcal{T}|$. Assign each colour c a distinct real character χ_c of $G = (\mathbb{Z}/2\mathbb{Z})^m$, with zero assigned the trivial character. Choose 2^m at least the number of colours. One G variable lies on each nearest-neighbour edge, shared by its endpoints. For a tile z at v, put

$$\mathcal{X}_{v,z}(g) = \prod_{e \ni v} \chi_{c(z,e)}(g_e).$$

Use $(p, q) = (1, 0)$ for the hard grammar and $(p, q) = (2, 1)$ for the soft grammar. Set $r = q/p$ and

$$c_z(s) = \begin{cases} q, & z \text{ marked and } s = -1, \\ p, & \text{otherwise,} \end{cases} \qquad F_v(g, s) = \sum_{z \in \mathcal{T}} c_z(s_v)\mathcal{X}_{v,z}(g), \quad L = pt.$$

Thus F_v is integral and $|F_v| \leq L$. Define

$$C_0 = 10000(L + 1), \quad D = C_0 + L, \qquad E_v = \mathbf{1}_{\{b_v > C_0 + F_v(g,s)\}}, \quad b_v \in \{1, \ldots, D\}. \tag{2.1}$$

The Hamiltonian sums these 0–1 terms and $20\mathbf{1}_{s_x \neq s_y}$ over bonds. Store east and north group variables at their left or lower endpoints. The alphabet is

$$\mathcal{A} = \{-1, 1\} \times \{1, \ldots, D\} \times G^2. \tag{2.2}$$

The site factor has support $\{v, v - e_1, v - e_2\}$; the bond terms have the two pair supports. This is a finite algorithmic table requiring no halting-time estimate. Every physical configuration has positive weight at finite temperature.

Summing the multiplicity variable gives exactly

$$\sum_{b_v=1}^{D} e^{-\beta E_v} = C_\beta(1+a_\beta F_v), \qquad C_\beta = C_0+Le^{-\beta}, \quad a_\beta = \frac{1-e^{-\beta}}{C_\beta}, \quad u = La_\beta < 10^{-4}. \tag{2.3}$$

The noninteger quantities here are derived weights, not microscopic couplings. Put $\lambda = pa_\beta$, so $t\lambda = u$.

2.4. **Exact cancellation and the positive gas.** For a finite set of factor centres V, integrate with normalized counting measure over every group edge incident to V, including dangling edges. Expand $\prod_{v\in V}(1 + a_\beta F_v)$. At each centre no tile or one tile is selected. Orthogonality gives

$$\int_G \chi_c\chi_d = \mathbf{1}_{c=d}, \qquad \int_G \chi_c = \mathbf{1}_{c=0}. \tag{2.4}$$

A selection survives exactly when internal colours match and dangling selected edges have colour zero. The grammar decomposes it into site-disjoint finite labelled rectangles γ. Denote sites by S_γ and marks by $M_\gamma \subset S_\gamma$. With $w_\gamma = \lambda^{|S_\gamma|}$, the exact identity is

$$\Xi_V(s) := \int \prod_{v\in V} (1 + a_\beta F_v(g, s))\, dg = \sum_{P:\, S_\gamma\subset V} \prod_{\gamma\in P} w_\gamma r^{N_-(M_\gamma,s)}. \tag{2.5}$$

The sum ranges over site-disjoint objects. For $r = 0$ use $0^0 = 1$: an occupied hard anchor requires plus spin. Every surviving term is nonnegative.

Without halting the hard table gives $\Xi_V = 1$, since only the empty selection survives. For the soft table all finite histories are unmarked, so Ξ_V is spin independent but generally not one. Both give exactly the bare Ising spin ratios. These partition identities must not be conflated.

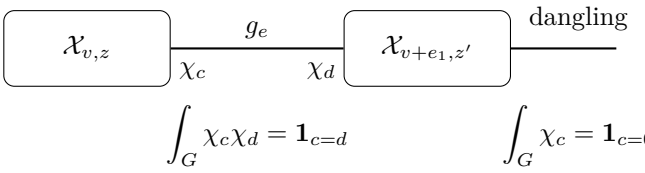


FIGURE 2. Shared group variables enforce matching, and freely summed dangling variables enforce zero exterior colour. Both are cancellation identities within a positive physical model.

Remark 2.2. Fixed dangling group values and periodic windings do not satisfy (2.5) in this form. We remove arbitrary physical group boundaries by conditional mixing before using free integration. Infinite matching tile planes are allowed by the grammar and are not classified by the finite-component lemma.

3. Canonical objects and projection of every Gibbs state

3.1. **A canonical gas without a cluster expansion.** Object species are labelled finite supports; incompatibility means overlap. Put $a_\gamma(s) = w_\gamma r^{N_-(M_\gamma,s)} \le w_\gamma$. Finitely many species lie in a fixed finite spatial volume. The number of size-n species containing a fixed site is at most $(16t)^n$: use a deterministic doubled spanning-tree walk to count the connected support, then label its sites. Thus rooted activity tails are uniformly summable. In particular,

$$B_{\mathrm{obj}} = 2\sup_x \sum_{\gamma:\, x\in S_\gamma} |S_\gamma| w_\gamma 2^{\operatorname{diam} S_\gamma} \le 2\sum_{n\ge1} n(32u)^n < 0.006442. \tag{3.1}$$

Lemma 3.1 (Finite-gas exchange)**.** *The free finite gases in* (2.5) *converge to a measurable canonical law Q_s, uniformly over complete spin profiles. Its decoded local patterns depend exponentially weakly on remote volume cutoffs and remote spin changes. It obeys all finite-species insertion and removal identities, and almost surely consists of finite objects. For each s it has a measurable coupling below an independent Bernoulli family (R_γ) with parameters w_γ, whose joint law is independent of s.*

Proof. Given all other species, occupation of γ has probability zero or $a_\gamma/(1+a_\gamma) \le w_\gamma$. Expose a finite gas sequentially using independent uniforms; the bound remains true after averaging over unexposed variables. This couples it below independent Bernoulli variables with parameters w_γ.

Compare two independent finite gases, retaining replica colours on occupied species. Join coloured species when their supports overlap. A component is blocked if it contains a species whose activities in the two gases differ, including a zero activity due to a cutoff. Exchange replica colours on every unblocked component meeting a query A. No exterior occupied species overlaps a moving object, all moving activities agree, and product weight and compatibility are preserved. The decoded query exchanges unless a component meeting it is blocked.

The two independent gases have independent upper families. A chain of distinct coloured species has occupation probability at most the product of upper activities. If it reaches distance R, the sum of support diameters is at least R. Summing over an initial site in A and successive overlap sites gives

$$\mathbb{P}(A \text{ reaches a discrepancy at distance } R) \le |A| \frac{B_{\mathrm{obj}}}{1-B_{\mathrm{obj}}} 2^{-R}. \tag{3.2}$$

Independence is used only for distinct species; allowing repetitions afterwards enlarges the numerical sum. No endpoint sum is needed. The estimate also covers one large object crossing the cutoff.

Local marginals are uniformly Cauchy, as are their remote-spin variations. Their consistent limits define Q_s. Rooted size tails ensure that every occupied label belongs to a finite species; mass cannot escape into an infinite component. Finite insertion/removal identities pass to cylinder tests after a size truncation, then to measurable tests by a monotone class. The occupation bound passes to the limit for every finite conditioning event. Enumerate the species as $\gamma_1, \gamma_2, \dots$ and sample their occupations successively using independent uniforms U_i. Each conditional probability given the preceding occupations is at most w_{γ_i}; on a zero-probability history set it to zero. Put $R_{\gamma_i} = \mathbf{1}_{U_i \le w_{\gamma_i}}$. Then the sampled gas lies below R, whose law is the fixed product law. The finite-prefix conditional probabilities are ratios of measurable cylinder probabilities of Q_s, so this is a measurable kernel in s. This proves all assertions. □

Blocked means different *activities*, not merely different availability. Soft activities can be unequal while both are positive. This finite exchange replaces the canonical-limit role of the cluster expansion. Loss networks give an established alternative; see appendix C.

3.2. **Uniform conditional mixing of the physical group variables.** After summing b, let $f_v = 1 + a_\beta F_v \in [1-u, 1+u]$. At fixed complete s, an edge has at most six other edges sharing an endpoint factor. Changing one such edge changes its unnormalised conditional density by a ratio in $[R^{-1}, R]$, where $R = (1+u)/(1-u)$. For a likelihood ratio in $[a,b]$, the total variation of the normalized laws is at most $(\sqrt{b}-\sqrt{a})/(\sqrt{b}+\sqrt{a})$. Here this is u. Thus the Dobrushin row sum satisfies

$$\alpha \le 6u < 0.0006. \tag{3.3}$$

Iteration of one-edge influence bounds a path of length at least d by $\sum_{n \ge d} \alpha^n$. It proves uniqueness of a conditional group law Q_s^g, exponential convergence uniformly over group

boundaries, exponential conditional covariance, and exponentially weak dependence on distant spins. The constants are independent of $|G|$ and s. These are classical Dobrushin comparison consequences, with exact applicability checked in appendix C.

3.3. **All-state projection and its inverse.** Physical Gibbs states exist: compact subsequential limits of finite-volume laws satisfy every fixed local DLR identity, since the alphabet is finite and finite-range kernels are continuous and positive. This also holds for bounded inhomogeneous fields.

Take any physical Gibbs state μ. After summing the b kernels, disintegration with respect to the complete spins makes the conditional group law a group Gibbs law, hence Q_s^g almost surely. If s, s' differ in finite Λ, put

$$L_{s,s'}(g) = \prod_{v\in\Lambda} \frac{f_v(g,s')}{f_v(g,s)} > 0.$$

Writing ΔE_K for the change of the Ising mismatch exponent, the spin marginal ν has conditional odds

$$\frac{\nu(s'_\Lambda \mid s_{\Lambda^c})}{\nu(s_\Lambda \mid s_{\Lambda^c})} = e^{-\Delta E_K + \sum_{x\in\Lambda} H_x(s'_x - s_x)} \mathbb{E}_{Q_s^g} L_{s,s'}. \tag{3.4}$$

All finite spin assignments have positive conditional probability. Uniform group mixing, *before* free integration, implies

$$\mathbb{E}_{Q_s^g} L_{s,s'} = \lim_{V\uparrow\mathbb{Z}^2} \frac{\Xi_V(s')}{\Xi_V(s)}. \tag{3.5}$$

The finite free-group expectation is exactly the ratio for $V \supset \Lambda$, and its local expectation converges uniformly. These limiting ratio kernels are consistent and quasilocal, either by this estimate or by lemma 3.1. There is no need to reconstruct an induced absolutely summable interaction.

Conversely, tilting Q_s^g by $L_{s,s'}$ gives a group Gibbs law at s'. Uniqueness gives

$$Q_{s'}^g = \frac{L_{s,s'} Q_s^g}{\mathbb{E}_{Q_s^g} L_{s,s'}}. \tag{3.6}$$

For a spin law obeying (3.4), its normaliser cancels the one in (3.6). Hence $\nu(ds)Q_s^g(dg)$ has the physical spin and group coordinate odds. Positive coordinate heat baths in any finite volume are irreducible and determine the joint kernel. Restore the independent local b conditionals to obtain the physical Gibbs lift. Projection and lift are inverse affine maps, with no invariance assumption. The kernel is translation covariant and depends only on the complete spins.

3.4. **A positive canonical lift.** Define $\rho(ds, dP) = \nu(ds)Q_s(dP)$. Fix exterior spins and take s^* plus inside finite Λ. If B is the minus set of another interior assignment, let

$$q_s(P) = r^{\sum_{\gamma\in P} |M_\gamma \cap B|}, \qquad b_s = \mathbb{E}_{Q_{s^*}} q_s.$$

For soft marks, $r^{|B|} \le q_s \le 1$. For hard marks this is the indicator of no anchor in B; its probability is positive by the summable vacancy product. Finite-volume identities and lemma 3.1 imply

$$Q_s = \frac{q_s Q_{s^*}}{b_s}, \qquad \lim_V \frac{\Xi_V(s)}{\Xi_V(s^*)} = b_s > 0. \tag{3.7}$$

The tilt is a decoded spatial cylinder, counting marks at sites of B even when objects cross Λ. The spin conditional in (3.4) is proportional to $e^{-E_K + \sum H_x s_x} b_s$. Multiplying by (3.7) cancels b_s exactly. Given the objects, spins have Ising conditionals with plus pins at hard anchors, or an additional field $-\frac{1}{2}\log r$ at soft marks. Given complete spins, objects have law Q_s.

Separate conditionals alone would not establish spatial DLR for tile labels. The next bridge includes all split and join edits.

Lemma 3.2 (Full spatial DLR for canonical lifts)**.** *Decode an object as its tile labels and put vacuum elsewhere. Every canonical lift satisfies the same nearest-neighbour spatial DLR specification: colour matching, occupied-tile weight λ, marked-minus weight r, and Ising bonds and fields. At $r = 0$ marked-minus configurations are forbidden. This concerns canonical finite-component lifts, not every possible hard-tile Gibbs law.*

Proof. A finite admissible edit in Λ meets finitely many old finite components. Every new component touching Λ lies in Λ together with finitely many old components incident to its boundary, hence is finite. Every admissible finite filling therefore preserves the sector.

Fix an initial and a final local spin-and-tile pattern. Remove all old objects meeting Λ, change spins in Λ, then insert all final objects meeting Λ. A removed exterior piece has a nonzero-edge connection toward Λ; unchanged exterior matching forces it into a final component meeting Λ. All temporary exterior changes are restored. After removal there is no hard anchor in Λ, so the intermediate spin change is permitted.

Partition by the finite lists of old and final labelled objects. There are countably many lists, and on each part the moves have fixed finite supports. Insertion/removal and conditional spin identities telescope to

$$e^{-\Delta E_K + \sum_{x\in\Lambda} H_x \Delta s_x} \lambda^{\Delta N_{\text{occupied}}} r^{\Delta N_{\text{marked},-}}. \tag{3.8}$$

For hard marks use admissible endpoints only; no zero weight is divided by zero. On each part the elementary identities give the corresponding change-of-measure identity for bounded tests. The replacement map is injective, since its initial local pattern is fixed and its exterior unchanged. Images of partition parts are disjoint. Summing over the lists gives the identity for the entire spatial replacement. These ratios identify the conditional law among every compatible filling, and are precisely the stated spatial DLR kernel. □

Remark 3.3. The tile weights λ, r are proof weights, not physical integer energies. The grammar can support infinite strips or planes and other hard-tile boundary sectors. No equality between that entire hard-tile simplex and the physical simplex is asserted.

4. A finite certificate selects a unique phase

4.1. **The geometric idea: cover differences, then exchange them.** On a halting input, the proof starts with two arbitrary candidate physical Gibbs states and passes to their canonical spin-and-tile lifts from section 3.4. Sample these two lifts independently. We want to show that their labels cannot sustain a disagreement that reaches infinity. This concerns the proof representation; the auxiliary group coordinates of the physical model are recovered separately.

First, a marked certificate will place the spin law of *every* candidate above the same zero-field Ising plus phase. The useful background is therefore plus spin and vacuum tile. A site that differs from this background either carries a minus spin or belongs to an occupied computation object. There are two ways to cover these exceptions in each replica. The whole support of each occupied object is covered by an upper object family R_i. Minus spins are covered by filled contour hulls of the lower Ising reference, in an upper contour family C_i. With replicas $i = 1, 2$, this gives four upper families.

The actual spins, objects and contours are correlated. The four *upper* families are independent Bernoulli fields indexed by possible species, not independent site labels. Their spatial supports can overlap, and contour hulls can be nested. Independence is obtained by a coupling whose upper object marginal stays fixed as the complete spins vary, followed by fresh randomness for the contour covering; section 4.5 proves the required joint independence. This distinction is what later permits multiplication of probabilities along a chain of distinct upper species.

A disagreement path must run through these covered exceptions. After adding a one-step collar, neighbouring exceptions belong to intersecting supports. A long disagreement

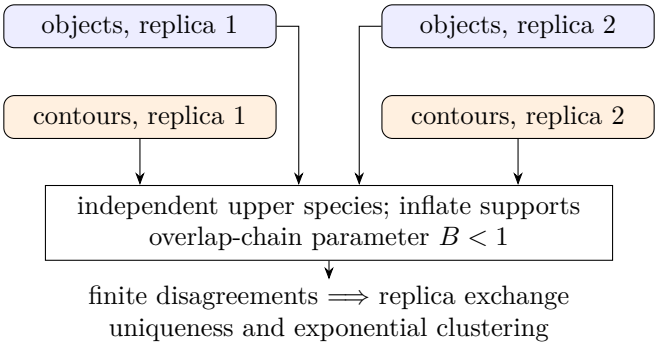


FIGURE 3. The four families are independent because each upper marginal is fixed before mixing over the spins and the reference coupling. Geometric supports may overlap and contour hulls may be nested.

therefore requires a long chain in the upper intersection graph. Small occupation probability at a typical site would not suffice: a rare very large support could connect many sites at once. The quantitative parameter below accounts for support size, overlap and diameter. Its bound below one makes every relevant chain component finite, with an exponential tail for its reach.

Now take the union K of disagreement components meeting a fixed local query. When K is finite, exchange the *entire* spin-and-tile labels of the two replicas on K. Along its exterior boundary the replicas agree. If W is any crossing-edge weight, a, b the two interior labels and τ their common exterior label, then

$$\underbrace{W(a,\tau)W(b,\tau)}_{\text{before exchange}} = \underbrace{W(b,\tau)W(a,\tau)}_{\text{after exchange}}.$$

Internal factors simply exchange between replicas as well. The product of local weights is unchanged, including hard colour constraints, so the common DLR specification makes the finite exchange probability preserving. It reverses the difference of the two query observables. That difference must have expectation zero, identifying the candidate laws. For two separated queries in one law, the same cancellation fails only when a disagreement component joins them; the exponentially small connection probability then gives clustering. Lemma 4.1 supplies the precise finite, random-region argument.

Thus the order of the proof is essential: a certificate selects the reference phase; independent coverings make disagreements finite; only then is the Gibbs exchange legitimate. Finite-box boundary convergence is a separate conclusion, proved in section 5.

4.2. **A uniform bias from one witnessed history.** In this subsection β is fixed; the bias h_* may depend on β and on the witnessed object. Let

$$\zeta = \sup_x \sum_{\gamma:\, x \in S_\gamma} w_\gamma < \infty.$$

The rooted count gives $\zeta \le 16u/(1-16u)$, which may replace ζ in the lower estimates below. Deleting additional species preserves the following elementary lower bound. For a finite S, sequentially require absence of all objects meeting S:

$$Q_s(\text{no object meets } S) \ge \prod_{\eta:\, S_\eta \cap S \neq \varnothing} (1+w_\eta)^{-1} \ge e^{-|S|\zeta}. \tag{4.1}$$

This uses the upper bound on conditional occupation, not stochastic monotonicity of the gas.

Fix spins other than s_x. In a finite gas, delete all species marked at x to form a baseline partition function Z_0. These deleted species are mutually incompatible, since all contain x.

Let

$$A = \sum_{\gamma:\, x\in M_\gamma} w_\gamma r^{N_-(M_\gamma\setminus\{x\},s)} Q_0(\text{no object meets } S_\gamma).$$

Then the two partition functions are $Z_+ = Z_0(1+A)$ and $Z_- = Z_0(1+rA)$. These formulas pass to the canonical ratios. If a marked species γ_* exists, let $n_* = |S_{\gamma_*}|$ and $k_* = |M_{\gamma_*}|$. Translate one chosen mark to x. In the soft case, and in the hard case with its single mark, respectively,

$$A \geq A_* := w_{\gamma_*} r^{k_*-1} e^{-n_*\zeta} > 0, \qquad h_* := \frac{1}{2}\log\frac{1+A_*}{1+rA_*} > 0, \tag{4.2}$$

where $r^0 = 1$ in the hard case. Translation makes this lower bound uniform over x. Consequently the spin single-site log odds are bounded below by

$$\log\frac{\nu(s_x = + \mid s_{x^c})}{\nu(s_x = - \mid s_{x^c})} \geq K\sum_{y\sim x} s_y + 2(H_x + h_*). \tag{4.3}$$

No quantitative lower bound on h_* independent of the computation is asserted. For an isolated one-mark object of weight w, with no competing objects, the spin partition ratio is simply $(1+w)/(1+rw)$, giving a field asymptotic to $(1-r)w/2$ as $w \downarrow 0$. The proof uses its strict positivity, not its magnitude.

4.3. **Comparison without target FKG.** Suppose $\inf_x H_x > -h_*$ and choose $0 < \delta \leq \inf_x(H_x + h_*)$. The right side of (4.3) is the log odds of a ferromagnetic Ising reference with uniform field δ. Condition the target on any exterior spins in a finite box, and compare to the reference with minus boundary. Couple single-site heat baths using the same uniforms. If the target configuration is above the reference, its update probability is at least the reference probability evaluated at the target, hence at least that evaluated at the lower reference. This preserves order. Finite positive heat baths converge to their unique stationary kernels. Passing to increasing boxes gives domination of the unique positive-field Ising law.

The reference's uniqueness at $\delta > 0$ is the Lebowitz–Martin-Löf/Lee–Yang consequence specified in appendix C. Decreasing the reference field to zero gives the zero-field plus law $\mu^+_{K,0}$. Thus every target spin law in the halting case satisfies

$$\nu \succeq \mu^+_{K,0}. \tag{4.4}$$

Only the reference is attractive. Target FKG is unnecessary and does not follow from positive object weights alone. For example, two incompatible objects marked separately at sites $1, 2$ give the factor $1 + wy_1 + wy_2$, with $y_i = 1$ for plus and r for minus. Its two-site lattice determinant equals $-w^2(1-r)^2 < 0$.

Uniformity of the field margin matters. Pointwise positive fields need not suffice: if $H_x > 0$ and $\sum_x H_x < \infty$, the bounded positive densities $\exp(\sum_x H_x s_x)$ tilt the two zero-field Ising phases into distinct Gibbs states for that inhomogeneous field. Their distinct tail events persist under equivalent bounded tilts.

4.4. **The reference contour covering.** In a finite plus box, extend all spins by plus. Disagreement bonds form finite even dual-edge graphs. Retain their connected components as graphs, including four-valent vertices, rather than resolving them into loops. Different components are vertex disjoint, and a component of n edges has weight e^{-Kn}. The correspondence with plus-exterior spins is exact; deletion of any one component remains admissible, including in nested configurations. Its conditional occupation is therefore at most $e^{-Kn}/(1+e^{-Kn})$, and sequential uniforms dominate the component gas by independent Bernoulli species of parameters e^{-Kn}.

Every minus site belongs to the filled hull of some occupied component. A hull of an n-edge component has at most $(n+1)^2$ sites and diameter at most n. The number of such

hull species covering a fixed site is at most

$$(2n+3)^2 16^n, \qquad n \geq 4. \tag{4.5}$$

To see this, choose a dual root within distance n and encode the connected edge set by a deterministic doubled-edge traversal of length $2n$. Filled hulls of different components may overlap; independence is asserted for the upper species, not for their geometric hulls. Uniform rooted contour tails permit passage to the infinite plus law: first truncate the contour size around a fixed query, pass to a coupled subsequential limit, then remove the truncation. This proves a coupling of $\mu^+_{K,0}$ to a fixed product contour covering without assuming an infinite-volume contour expansion.

4.5. **Why there really are four independent families.** For each canonical target law couple $s \geq \eta$ with $\eta \sim \mu^+_{K,0}$. Conditional on s, sample the objects together with their product upper family R using a kernel depending on s alone. Its R marginal is a fixed product law, independent of s. It follows that R is independent of the entire pair (s, η): for bounded F and an event E,

$$\mathbb{E}[F(s,\eta)\mathbf{1}_{R\in E}] = \mathbb{P}(R \in E)\mathbb{E}F(s,\eta).$$

Now draw the upper contour family C conditionally on η, using fresh randomness from the reference coupling. This preserves independence of R and C. Since $s \geq \eta$, the minus sites of s are covered by C. The objects are covered by R. Repeat the entire construction independently for two independent canonical target states. The resulting families R_1, C_1, R_2, C_2 are mutually independent. Merely knowing conditional stochastic domination would not prove this independence.

Inflate each object support and each filled hull by one nearest-neighbour collar. Put a vertex in a graph for every occupied upper species, with edges for intersecting inflated supports. If two canonical replicas disagree at a site, that site is minus in at least one replica or occupied by an object in at least one replica. Every nearest-neighbour disagreement path is therefore covered by an *intersection chain* of inflated supports. This does not identify the intersection graph with the nearest-neighbour components of the union of supports.

For all four families define

$$B = \sup_x \sum_{\xi:\, x\in S^+_\xi} |S^+_\xi| z_\xi 2^{\operatorname{diam} S^+_\xi}, \tag{4.6}$$

where z_ξ is the upper occupation parameter. The elementary count gives

$$\begin{aligned} B &\leq 200 \sum_{n\geq 1} n(32u)^n + 200 \sum_{n\geq 4} (n+1)^2(2n+3)^2(32e^{-K})^n \\ &< \frac{2}{3} \qquad (K \geq 8,\ u < 10^{-4}). \end{aligned} \tag{4.7}$$

The numerical derivation is in appendix B. It applies to the soft table for all $\beta \geq 2/5$ and to the hard single-anchor table at $\beta = 1$.

A chain reaching distance R from A has total support diameter at least R. Apply independence to distinct coloured species, sum over overlap sites, then enlarge to all numerical chains. We obtain

$$\mathbb{P}(A \text{ has a covering arm to distance } R) \leq |A| \frac{B}{1-B} 2^{-R} < 2|A|2^{-R}, \tag{4.8}$$

and, in particular,

$$\mathbb{P}(A \leftrightarrow B \text{ in disagreements}) \leq 2|A||B|2^{-\operatorname{dist}(A,B)}. \tag{4.9}$$

The size tails and the arm bound make every relevant occupied intersection component finite, both spatially and in number of species. Hence all site-disagreement components are finite.

4.6. **The finite exchange principle.** We state explicitly the established mechanism being used.

Lemma 4.1 (Independent-disagreement exchange)**.** *Let two laws on a finite alphabet satisfy the same nearest-neighbour DLR specification, possibly with hard constraints. If their independent product has only finite nearest-neighbour disagreement components, the laws are equal. For two copies of one such law, and local f, g on A, B,*

$$|\operatorname{Cov}(f,g)| \le 2\|f\|_\infty \|g\|_\infty \mathbb{P}(A \leftrightarrow B \text{ in disagreements}).$$

Proof. Swap replicas on the union D_A of disagreement components meeting A. It is finite. Exterior labels on its boundary agree in the replicas. Internal factors exchange, and the product of crossing-edge factors is unchanged, including when a factor is zero. The disagreement set and the choice of D_A are invariant under the swap.

For each deterministic finite set D, the product DLR conditional on its exterior is invariant under swapping on the event $D_A = D$. The event is determined by its interior disagreements and agreeing boundary, and is invariant under the same operation. Partition over the countably many finite D to justify the random finite swap. It reverses $f_1 - f_2$, proving equality of expectations for any local f.

For covariance, use

$$\operatorname{Cov}(f,g) = \tfrac{1}{2}\mathbb{E}[(f_1 - f_2)(g_1 - g_2)].$$

On the invariant event that no disagreement component joins A to B, the swap reverses the integrand. Its expectation there is zero. On the complement its magnitude is at most $4\|f\|_\infty\|g\|_\infty$, giving the bound. □

This is the cluster-exchange argument of [22], here proved for the canonical constrained representation. The proof also gives a reusable criterion: any locally decoded finite-object representation with a uniform lower Ising comparison and independent upper object and contour families satisfying (4.6) with $B < 1$ has at most one canonical spatial Gibbs law, and covariance bounded by $2B/(1-B)$ times the corresponding exponential chain bound. Rectangular computation histories are one way to verify these hypotheses; the exchange itself is not specific to computation.

By lemma 3.2, all canonical lifts of arbitrary physical Gibbs states satisfy the same specification. Equations (4.8)–(4.9) and lemma 4.1 make these lifts equal. Projection and the unique physical lift then prove full physical uniqueness, not merely translation-invariant uniqueness. They also prove (1.1), and the same estimate for local spin-and-tile observables.

4.7. **Transfer to every physical observable.** Let F, G be bounded physical observables on finite sets A, B. Conditional on the complete spins,

$$\operatorname{Cov}_\mu(F,G) = \mathbb{E}_\nu \operatorname{Cov}(F,G \mid s) + \operatorname{Cov}_\nu(\overline{F}(s), \overline{G}(s)), \qquad \overline{F} = \mathbb{E}(F \mid s). \tag{4.10}$$

The b variables have independent local conditionals; summing them enlarges the group support by a fixed collar. The first term is exponentially small by (3.3). The same influence-path estimate shows that replacing all spins outside A^ℓ by plus changes $\overline{F}$ by at most

$$C_A\|F\|_\infty \frac{\alpha^{\ell-2}}{1-\alpha}, \tag{4.11}$$

with an effectively bounded C_A, and analogously for G. The truncated conditional means are bounded spin functions. Apply (1.1) to them, with $\ell = \lfloor \operatorname{dist}(A,B)/4 \rfloor$. Their supports grow polynomially in ℓ while their separation is at least half the original distance. Absorbing that polynomial gives

$$|\operatorname{Cov}_\mu(F,G)| \le C_{A,B}\|F\|_\infty\|G\|_\infty e^{-c\operatorname{dist}(A,B)}, \qquad c = \tfrac{1}{4}\log 2, \tag{4.12}$$

with computable support-dependent constants, uniform over inputs and the stated temperatures. Small distances are absorbed by increasing $C_{A,B}$. This proves clustering for the entire physical alphabet, not only for its spin coordinate.

4.8. **The full nonhalting simplex and undecidability.** Without halting, (2.5) is spin independent. All ratios in (3.5) are one, so projection and lift identify exactly all zero-field Ising Gibbs laws. At $K \geq 8$ the reference is below its critical temperature. An elementary Peierls bound already gives coexistence at $K = 7$: using (4.5), its minority bound is less than $121/3072 < 1/2$. Hence $J_c \leq 7/2 < K/2$; the exact critical value is unnecessary here.

Coquille–Velenik's Aizenman–Higuchi corollary now applies to this exact two-dimensional nearest-neighbour zero-field Ising specification. Every Ising Gibbs state is $\theta\mu^+_{K,0}+(1-\theta)\mu^-_{K,0}$, $0 \leq \theta \leq 1$. The affine physical lift preserves decompositions and extremality. Its translation covariance proves that every physical Gibbs state is translation invariant. There are exactly two extremal physical states $\mu^{\pm}_M$.

The plus endpoint has the four-family covering by taking the spin reference itself; the minus endpoint follows by spin reversal of the Ising marginal and the same auxiliary transfer. Thus both cluster for every physical observable. Put $m_K = \mu^+_{K,0}(s_0)$. The contour estimate gives $m_K > 0.99$ (indeed the minority probability is less than 10^{-6}). In an interior mixture,

$$\mathrm{Cov}(s_0, s_x) = \mathrm{Cov}_{\mu^+_{K,0}}(s_0, s_x) + 4\theta(1-\theta)m_K^2. \tag{4.13}$$

The second term is nonzero and the first decays. For the equal mixture, ferromagnetic association makes the covariance at every separation at least $m_K^2 > 0.99$. At $K = 20$ the hard construction has the stronger minority bound recorded in appendix B.

Halting is equivalent to a singleton physical simplex by the preceding proof. Therefore a uniqueness decision algorithm for the exact tables would decide halting. On the machine-indexed promised family the uniqueness predicate is Σ^0_1-complete and coexistence is Π^0_1-complete. These are classifications of this effective family, not upper classifications for every finite-range interaction.

5. Finite-volume convergence and its effective scope

5.1. **A uniformly computable selected phase.** For a box V, define a spin-and-object law ρ^+_V with all exterior spins plus, vacuum exterior tiles, and objects contained in V. Its spin weight is the Ising plus-boundary weight times $\Xi_V(s)$. The latter is coordinatewise nondecreasing in the spins, whether or not a marked object exists. The common-uniform heat-bath argument therefore compares the finite spin law from below to the Ising law with the *same plus boundary*. No positive h_* is needed. This remains true for nonnegative inhomogeneous fields.

The two finite contour coverings and two object coverings have the bounds of section 4 uniformly in the volume. Take independent laws in nested boxes $V \subset W$, extending their spin-and-tile configurations by plus and vacuum. They obey the same local specification inside V. Exchange disagreement components meeting $A \subset V$ whenever they do not reach V^c. Product DLR makes the exchange valid, as in lemma 4.1. Failure requires a covering arm to V^c. Thus, for local f on A,

$$|\rho^+_V(f) - \rho^+_W(f)| \leq 2\|f\|_\infty |A| \frac{B}{1-B} 2^{-\operatorname{dist}(A,V^c)} < 4\|f\|_\infty |A| 2^{-\operatorname{dist}(A,V^c)}. \tag{5.1}$$

It follows that ρ^+_V converges exponentially to a selected canonical law ρ^+. The same comparison for displaced boxes gives translation invariance at uniform field. The replica covariance argument gives clustering. In the nonhalting case its spin law is precisely the Ising plus phase; in the halting case it is the unique spin law.

There is a physical version of this sequence. Include the site factors centred in V, sum their b variables and *all* group variables on edges incident to V, and impose plus spins outside V for the Ising bonds. By (2.5) its spin marginal is exactly that of ρ^+_V. Deep in V its conditional group kernel converges uniformly to Q^g_s. The decomposition into conditional means in section 4.7, now using (5.1) and truncation at one quarter of the boundary distance, gives an exponential rate for every fixed local physical observable. All constants can be

bounded effectively from its support using (3.3). Each limiting local physical DLR identity is valid, so the limit is the physical lift μ_M^+.

For computable β, the model table and any finite physical expectation are computable by finite enumeration. Given an observable and error, the preceding uniform rate chooses a sufficient box. Thus μ_M^+ is computable uniformly in the machine input, including nonhalting inputs. The required enumeration is enormous and no efficient sampling algorithm is asserted. The free incident group variables are part of the boundary convention; (5.1) is not an estimate for every fixed microscopic boundary.

5.2. **Arbitrary spin boundaries on the halting branch.** A stronger, but still qualified, statement is available. Let γ_* be a witnessed marked history and choose d_* larger than its diameter plus a fixed collar. In any free-group box V, the translated witness lies inside V whenever x is at distance greater than d_* from V^c. The finite-volume vacancy argument then gives the same h_* at every such x, for arbitrary exterior spins. Set $U = \{x \in V : \operatorname{dist}(x, V^c) > d_*\}$. Conditioning on the spins in $V \setminus U$, the spin law on U dominates the Ising reference with field h_* and minus U boundary.

The following exact reference input is used: Ott [20, Theorem 6.1, arXiv version 2] bounds the centre plus/minus magnetization difference in a box of radius N, at fixed ferromagnetic coupling and uniform positive field, by $C_0 e^{-\nu_0 N}$. Our reference coupling is $K/2$ and field $h_* > 0$. Domain monotonicity and an inner box centred at x imply an ordered coupling of its finite-minus law η_U^- and its infinite positive-field law η_h with

$$\mathbb{P}(\eta_U^-(x) \neq \eta_h(x)) \leq C_0 e^{-\nu_0 \operatorname{dist}(x, U^c)}. \tag{5.2}$$

Absorb the factor $1/2$ converting magnetization to mismatch into C_0. Glue this to the target domination and to $\eta_h \geq \eta_0^+$. Off the mismatch set, a target minus site is covered by a contour of η_0^+. The object upper marginal remains independent of the entire reference chain by the constant-marginal construction in section 4.5.

For two boxes with arbitrary spin boundaries, let $D = \operatorname{dist}(A, V^c)$ in their common inner box and put $q = D - d_* > 0$, $R = \lfloor q/2 \rfloor$. For any $x \in A$, a disagreement arm to the boundary either has a four-family covering arm of length R, or encounters a reference mismatch in the radius-R buffer about x. A union bound over this buffer, not over the whole boundary, gives

$$\mathbb{P}(A \text{ has a disagreement arm to } V^c) \leq |A| \left[\frac{B}{1-B} 2^{-R} + 2C_0 (2R+1)^2 e^{-\nu_0 (q-R)} \right]. \tag{5.3}$$

For the initial arm, all its covered sites preceding the first mismatch form a chain; thus a mismatch encountered after that chain leaves the buffer is already counted by the first term. Finite exchange inside the common box now proves exponential convergence uniformly over spin boundaries. Transfer to physical observables uses conditional group mixing exactly as before.

The constants in this conclusion depend on the witnessed history, K, and the positive-field reference constants. We have not extracted numerical C_0, ν_0. This proves existence of a model-dependent exponential rate for *free incident group variables and arbitrary spin boundaries*. It is not arbitrary microscopic-boundary mixing or strong spatial mixing.

5.3. **A general effective compactness theorem.** There is a useful general limitation on what "noncomputable convergence despite uniqueness" can mean.

Theorem 5.1 (A valid modulus under the uniqueness promise)**.** *Let a finite-alphabet finite-range finite-energy specification on $\mathbb{Z}^d$ have computable local kernels, with its finite alphabet and a finite-range bound supplied effectively. If it has a unique Gibbs state, that state and a valid local all-boundary convergence modulus are computable. One uniform partial algorithm terminates on every such unique input. The assertion includes computable translation-invariant finite tables at computable temperature.*

Proof. Fix a finite observable support A and nested boxes $V_n \supset A$. Finite range means that the marginal kernel on A depends on the exterior only through a finite boundary collar. Define the computable real

$$D_n(A) = \max_{\tau,\tau'} \mathrm{TV}\big((\gamma^{\tau}_{V_n})_A, (\gamma^{\tau'}_{V_n})_A\big).$$

The maximum is over finitely many collar assignments and consists of finite positive sums. If $D_n(A)$ does not tend to zero, choose boundary pairs separated by a fixed amount along a subsequence. Compactness gives two subsequential limiting laws. Every fixed DLR identity passes to these limits by locality; they are Gibbs states with different A marginals, contradicting uniqueness.

Search over n and rational precision for a strict certificate $D_n(A) < \varepsilon$. Approximation to the finitely many probabilities semidecides this strict inequality. Under uniqueness the search terminates. Its certificate is valid for every larger box and every Gibbs law: by conditioning on the collar of V_n, any such A marginal is a convex mixture of the inner kernels whose diameter was bounded. The same is true for every larger-volume boundary assignment. Hence any one inner kernel approximates the unique marginal, and the certificate controls convergence for all larger boxes, not just the first one found.

Dovetail this construction over finite supports and errors 2^{-k}. It computes both the state and a valid modulus. No proof of uniqueness is required as input; termination uses the uniqueness promise. □

For general computable finite-range specifications, this also gives a Π^0_2 upper description of uniqueness: for each finite support in an exhaustion and each k, there is a finite strict-diameter certificate below 2^{-k}. The strict numerical certificate has a decidable finite witness. No completeness claim for this general upper bound is made.

Proposition 5.2 (No total family-wide minus-boundary bound)**.** *There is no total computable function of the machine input returning a box size beyond which the free-group, minus-spin finite-volume mean is within* $1/4$ *of its infinite-volume value on every halting input. The function is not required to be valid on nonhalting inputs. The same impossibility holds for an all-boundary bound.*

Proof. On every nonhalting input, the finite spin law is exactly minus-boundary zero-field Ising and has centre mean less than -0.99 at every box size, by reference monotonicity. On every halting input the unique infinite-volume mean exceeds 0.99, by (4.4). If the total function existed, evaluate the finite mean at its output size to error $1/8$. The result is negative on every nonhalting input and positive on every halting input. This decides halting, a contradiction. For a bound stated for actual physical DLR boxes, enlarge the returned box by a fixed collar of width two. The natural law then contains every interaction meeting the inner box; conditioning on its physical collar gives the exact inner DLR kernel. Its query marginal is a mixture of those kernels and obeys the all-boundary certificate. The same finite computation again decides halting. □

Thus a fixed unique computable model does not furnish a noncomputable valid convergence modulus. The obstruction is to a total rule across this undecidable family. The plus sequence (5.1) is a positive exception precisely because it selects an equilibrium on both branches.

6. Pressure, smooth response and one-sided computability

For a uniform dimensionless field h, define the physical pressure using counting measure on the full alphabet:

$$p_\beta(h) = \lim_{L\to\infty} \frac{1}{L^2} \log Z^{\mathrm{free}}_{[1,L]^2}(h),$$

where the free Hamiltonian includes terms wholly contained in the box. Finite range gives existence and independence of ordinary boundary conventions. Normalized Haar

integration used earlier changes only explicit spin-independent normalising constants, never the derivatives in h.

6.1. **Pressure values and slopes.** For our nonnegative physical terms, tiling large boxes by L-boxes and deleting crossing terms gives

$$p_L(h) - \frac{42\beta}{L} \le p_\beta(h) \le p_L(h), \qquad p_L(h) = L^{-2} \log Z^{\mathrm{free}}_{[1,L]^2}(h). \tag{6.1}$$

Indeed at most $2L$ site factors and $2L$ mismatch bonds per block cross the tiling; their energies are at most 1 and 20. Remainders vanish in the thermodynamic limit. The field terms are on-site and create no crossing error. Finite positive partition sums are computable, so (6.1) computes the pressure uniformly for computable β, h. This elementary estimate is a special case of established pressure computability results; it is not an undecidability of the pressure value.

The finite-alphabet full-shift variational principle identifies translation-invariant Gibbs laws with equilibrium laws for finite-range interactions; precise hypotheses are checked in appendix C. Consequently

$$p'_\beta(h-) = \min_{\mu \in \mathcal{G}_{\mathrm{TI}}(h)} \mu(s_0), \qquad p'_\beta(h+) = \max_{\mu \in \mathcal{G}_{\mathrm{TI}}(h)} \mu(s_0). \tag{6.2}$$

For completeness, pressure is the supremum of the affine functions $\mathsf{h}(\mu) - \beta e(\mu) + h\mu(s_0)$ on the compact invariant measure space. Entropy is upper semicontinuous on a finite full shift and energy is continuous. Taking maximisers at $h + t$ along $t \downarrow 0$ gives a maximiser at h; comparing the two variational inequalities identifies the right derivative as the maximum slope of these maximisers. The left derivative is the minimum, by the same argument.

On the nonhalting branch, (2.5) implies

$$p_\beta(h) = c_\beta + p^{\mathrm{Ising}}_K(h)$$

with a spin-independent c_β. In the soft case this includes the nontrivial unmarked-gas pressure. Hence $p'_\beta(0\pm) = \pm m_K$. On the halting branch the unique state has positive magnetization exceeding 0.99, so both slopes coincide and are positive. Thus differentiability at zero is equivalent to halting.

6.2. **Fixed-order cumulants from local clustering.** For bounded random variables $X_1, \dots, X_n$, write

$$\kappa_n(X_1, \dots, X_n) = \sum_{\pi \in \mathcal{P}_n} (|\pi| - 1)!(-1)^{|\pi|-1} \prod_{C \in \pi} \mathbb{E} \prod_{i \in C} X_i,$$

where $\mathcal{P}_n$ is the set of partitions of $\{1, \dots, n\}$.

Lemma 6.1 (Summable cumulants)**.** *Any class of spin laws satisfying* (1.1) *uniformly obeys, for $n \ge 2$,*

$$|\kappa_n(s_{x_1}, \dots, s_{x_n})| \le C_n 2^{-\operatorname{diam}\{x_1,\dots,x_n\}/(n-1)}, \qquad C_n = 4n^2 \sum_{\pi \in \mathcal{P}_n} |\pi|!. \tag{6.3}$$

Every fixed-order cumulant is absolutely summable over relative positions, with a computable uniform tail.

Proof. For positive diameter, remove a longest edge from a minimum spanning tree on the indexed positions. Its length a is at least the diameter divided by $n - 1$. The resulting parts I, J have every cross-distance at least a, since a shorter crossing edge would improve the tree. Replace the joint law by the product of its I and J marginals. Its mixed cumulant is zero.

In the moment formula, a block meeting both parts changes by a covariance of two spin products, of magnitude at most $4n^2 2^{-a}$ by (1.1); noncrossing blocks do not change. Every moment has magnitude at most one. Telescoping products for a partition with $|\pi|$

blocks, then multiplying by $(|\pi|-1)!$, yields (6.3). Repeated positions cannot lie across a positive-distance cut. At diameter zero the elementary moment bound is smaller than C_n.

With $x_1 = 0$, tuples whose maximum radius is m number at most $(2m+1)^{2(n-1)}$. An explicit tail beyond radius R is therefore

$$C_n \sum_{m \geq R} (2m+1)^{2(n-1)} 2^{-m/(n-1)}. \tag{6.4}$$

For each fixed n this converges geometrically after a computable initial segment, uniformly over the class. □

6.3. **The inhomogeneous-field bridge to response.** Infinite-volume clustering alone does not justify differentiating a thermodynamic limit. Here the missing uniformity follows from the proof: on a halting input, uniqueness and (1.1) hold for *every bounded field profile* with infimum greater than $-h_*$. This includes arbitrary finite field tilts, with the same covariance and cumulant constants.

Proposition 6.2 (Smooth pressure)**.** *On a halting input, for either stated construction and temperature, $p_\beta \in C^\infty((-h_*, \infty))$. For $n \geq 2$,*

$$p'_\beta(h) = \mu_h(s_0), \qquad p^{(n)}_\beta(h) = \sum_{x_2,\ldots,x_n \in \mathbb{Z}^2} \kappa_{n,h}(s_0, s_{x_2}, \ldots, s_{x_n}). \tag{6.5}$$

The sums converge absolutely and locally uniformly. In particular $0 \leq p''_\beta(h) \leq 68$.

Proof. For finite Λ let $\mu_{\Lambda,t}$ be the probability law with density proportional to $\exp(t\sum_{x\in\Lambda} s_x)$ relative to μ_h. The local DLR equations identify it as a Gibbs law with field $h + t\mathbf{1}_\Lambda$. Uniqueness identifies it with the unique law for that profile. Finite differentiation gives, for local F,

$$\frac{d}{dt}\mu_{\Lambda,t}(F) = \sum_{x\in\Lambda} \mathrm{Cov}_{\mu_{\Lambda,t}}(F, s_x). \tag{6.6}$$

On a compact t-interval inside the permitted range, local subsequential limits as $\Lambda \uparrow \mathbb{Z}^2$ satisfy the uniform-field DLR equations and equal μ_{h+t}. Convergence is uniform in t: otherwise take a convergent sequence of parameters witnessing failure and apply the same compactness argument. Uniform clustering bounds the sum tails in (6.6). Passing to the integrated identity proves differentiability of local moments and its infinite response sum.

For higher orders, finite-tilt differentiation of a cumulant adds one spin variable summed over Λ. Apply lemma 6.1 at each fixed order. Uniform local moment convergence and (6.4) justify passage through the integrated identity, including the sums in all relative positions. Induction with the fundamental theorem of calculus proves every derivative in (6.5); it does not formally differentiate an uncontrolled infinite series. The first identity follows from (6.2) and uniqueness. Finally convexity gives $p'' \geq 0$, and

$$p''(h) \leq 4 \sum_{x\in\mathbb{Z}^2} 2^{-\|x\|_\infty} = 4\left(1 + 8\sum_{m\geq 1} m 2^{-m}\right) = 68.$$

□

This is a concrete use of the classical cumulant-response mechanism, rather than a new general implication from clustering. The available C_n do not imply an $n!C^n$ derivative bound. We assert no analyticity or convergent Taylor series for the target model.

6.4. **The computable right jet and the noncomputable left slope.** For every input, including nonhalting ones, all bounded field profiles with positive infimum give uniqueness: use (4.3) with $h_* = 0$ if no certificate is available. The clustering estimates are uniform as the positive uniform field decreases to zero. The selected finite-volume estimate (5.1) is also uniform for $h \geq 0$. Since finite-volume probabilities are continuous in h, their uniform limit μ_h^+ is locally right-continuous at zero. Uniform cumulant tails give the same continuity for each sum in (6.5).

Apply the finite-tilt response argument on $h > 0$ and pass through its integrated identities to the endpoint. Every right derivative of pressure at zero exists, and equals the cumulant sum of the selected zero-field plus state (with $n = 1$ interpreted as magnetization). This is a smooth one-sided jet; it need not determine pressure on any neighbourhood.

The jet is uniformly computable. For order $n \geq 2$, choose a radius using (6.4); enumerate the finitely many remaining tuples and partitions; compute their selected local moments with (5.1) to sufficient total accuracy. The cases $n = 1$ and $n = 0$ use the selected mean and (6.1). All choices terminate without deciding halting.

In contrast, (6.2) gives

$$p'_\beta(0-) = \begin{cases} -m_K < -0.99, & \text{without halting,} \\ \mu_M(s_0) > 0.99, & \text{with halting.} \end{cases}$$

A uniform computation of this real number even to error $1/4$ would decide halting by its sign. This is a family-wise noncomputability statement, compatible with computability of each pressure value, the selected phase, its complete right jet, and the valid modulus under a uniqueness promise.

7. Perturbations and exact restoration of coexistence

7.1. **Spin-only multibody perturbations.** Add to the exponent a finite-range interaction $\sum_X \Psi_X(s_X)$; translation invariance and ferromagnetism are not required. Put

$$d_X = \operatorname{osc}(\Psi_X), \qquad \delta = \sup_x \sum_{X \ni x} d_X, \qquad p_X = 1 - e^{-d_X},$$

and, with X^+ the nearest-neighbour collar, define the two-replica activation parameter

$$B_\Psi = 2 \sup_x \sum_{X\colon x \in X^+} |X^+| p_X 2^{\operatorname{diam} X^+}. \tag{7.1}$$

Use B_0 for the unperturbed bound in (4.7).

Theorem 7.1 (A spin perturbation neighbourhood)**.** *On a halting input, at a uniform field h, suppose*

$$h + h_* > \delta/2, \qquad B_0 + B_\Psi < 1. \tag{7.2}$$

Then the perturbed physical model has exactly one Gibbs state, exponentially clustering for every bounded local physical observable. For spin observables its covariance is bounded by

$$\frac{2B}{1-B} \|f\|_\infty \|g\|_\infty |A||C| 2^{-\operatorname{dist}(A,C)}, \qquad B = B_0 + B_\Psi.$$

If $\operatorname{diam} X \leq R$*, a sufficient condition is*

$$\delta < \min\{h + h_*, (6C_R)^{-1}\}, \quad C_R = 10 v_R 2^{R+2}, \quad v_R = \min\{5(R+1)^2, (R+3)^2\}. \tag{7.3}$$

These are perturbations of the distinguished spins only.

Proof. A one-site log odds changes by at most $\sum_{X \ni x} d_X \leq \delta$. Thus the lower reference field is $h + h_* - \delta/2 > 0$. Comparison with the zero-field plus reference still holds, without any sign assumption on Ψ.

For each finite support, write the positive expansion

$$e^{\Psi_X(s_X)} = e^{\min \Psi_X} [1 + q_X(s_X)], \qquad q_X = e^{\Psi_X - \min \Psi_X} - 1 \geq 0. \tag{7.4}$$

Conditional on the spins, introduce independent activation variables with probability $q_X/(1+q_X) \le p_X$. Multiplying the spin weight by their conditional kernels cancels the factors $1+q_X$: the augmented weight has factor $q_X(s_X)$ when active and 1 when inactive, apart from a spin-independent constant. This cancellation remains valid where $q_X = 0$; no strictly positive likelihood ratio between activation laws is presumed.

Use fresh uniforms for a product activation upper family whose law, conditional on the complete spins and all previous variables, is fixed with parameters p_X. It is therefore independent of the four families in section 4.5. Two replicas add two independent families. Their inflated intersection parameter is (7.1), so the same chain estimate holds with $B = B_0 + B_\Psi < 1$.

For exchange, include the entire support of every occupied object and every active factor in the actual bad hypercomponents, together with minus sites in both replicas, joining neighbouring sites and incident supports. Entire active supports must be used even if disconnected. Each such component lies in a finite component of the six upper families. Swap spins, whole objects and activation labels on the finite union of components meeting the query. No object or active factor crosses its boundary. Every inactive crossing factor contributes 1, and an internal active factor exchanges its full spin pattern and activation label. Thus augmented product weights, including zeros, are preserved. One may verify the same identity by finite removal, spin change and reinsertion, as in lemma 3.2; this avoids dividing by a zero activation weight. Partition over the finite regions and finite object/factor lists to justify the random exchange. The covariance and uniqueness proof of lemma 4.1 now applies verbatim to these hypercomponents. Physical group conditionals remain unchanged, so section 4.7 transfers clustering.

Finally $|X^+| \le v_R$, $\operatorname{diam} X^+ \le R+2$, and $x \in X^+$ implies X meets the five-site nearest-neighbour neighbourhood of x. Since $p_X \le d_X$, this gives $B_\Psi \le C_R\delta$. Under (7.3), $B < 2/3 + 1/6 = 5/6$ and the field margin remains positive. □

In physical energy units $\Psi = -\beta W$. For each fixed range and each witnessed halting input this gives an explicit open spin neighbourhood. It does not control perturbations of the multiplicity or group tables.

7.2. **Arbitrarily small microscopic symmetrization.** The preceding scope cannot simply be replaced by a uniform radius for arbitrary physical perturbations.

Proposition 7.2 (No family-wide real-potential radius)**.** *At $\beta = 1$, for every $\varepsilon > 0$ there is a halting input whose unperturbed physical interaction, in either construction, admits a translation-invariant real-valued perturbation on the same supports of norm below ε with at least two Gibbs states. Here the norm is* $\sup_x \sum_{X \ni x} \|W_X\|_\infty$.

Proof. For one site let $U = \sum_{z \text{ unmarked}} \mathcal{X}_z$ and $V = \sum_{z \text{ marked}} \mathcal{X}_z$, with m marked tile types. The two summed factors are

$$f_+ = 1 + a_\beta p(U+V), \qquad f_- = 1 + a_\beta(pU + qV).$$

At $\beta = 1$, put $f_{\mathrm{sym}} = (f_+ + f_-)/2$ and add the b-independent physical site energy

$$\Delta(s,g) = \log \frac{f_s(g)}{f_{\mathrm{sym}}(g)}. \tag{7.5}$$

Its multiplicity sum is exactly $C_\beta f_{\mathrm{sym}}$, independent of s. The group factor stays in $[1-u, 1+u]$ and has the same Dobrushin estimate. The summed system is therefore the product of a unique group law and zero-field Ising. Lift either Ising phase and restore the local multiplicity kernels to obtain distinct physical Gibbs states. Moreover,

$$\|\Delta\|_\infty \le \frac{a_\beta(p-q)m}{2(1-u)}. \tag{7.6}$$

Each site lies in at most three translated site-factor supports; the interaction norm is at most three times this quantity.

It remains to make m/t small while keeping halting. Start with a one-step-halting machine whose initial state is not halting, and add unreachable states. Let P be the number of ordered tape-symbol pairs and Q the number of machine states. For either permanent-bit value the folded cell alphabet has

$$\sigma_0 = P(1 + 2Q) + 1, \qquad \sigma = 2\sigma_0$$

letters, including poison. Interior-row, interior-column tiles alone give $t \geq \sigma^2\sigma_0$: their neighbour letters are arbitrary, their centre has bit zero, and their visible edge data distinguish the choices. For the hard grammar the marked types number $m = 3P\sigma$, whence

$$\frac{m}{t} \leq \frac{3}{8PQ^2} \longrightarrow 0.$$

For the soft grammar, bottom rows have no halting centre for this chosen initial word. In the other two row types, the possible halting centres and column statuses give the safe bound

$$m \leq 4P\sigma^2 + 8P\sigma, \qquad \frac{m}{t} \leq \frac{8P}{\sigma} + \frac{16P}{\sigma^2} \longrightarrow 0.$$

These counts follow directly from the explicit tuples in appendix A; in particular they survive deduplication of identical visible tile data. Padding changes no reachable transition and thus no genuine history. Since $a_\beta < 1/[10000(pt + 1)]$, (7.6) tends effectively to zero. Choose enough unreachable states to make the interaction norm less than ε. □

The perturbations are finite computable *real* tables. A constant shift can make them nonnegative without changing Gibbs laws, at the cost of at most a factor two in the norm bound. This result does not concern a topology restricted to integer-valued tables, and it does not exclude a physical perturbation neighbourhood for each fixed halting input. Nor does it establish a first-order transition under a small uniform negative field: selection of plus at one field and minus at another would not by itself prove coexistence between them.

8. Scope and further questions

The construction separates three questions that can otherwise look similar. A selected clustered phase is uniformly computable. Uniqueness among all Gibbs states is undecidable. Under a uniqueness promise, a valid all-boundary convergence modulus is computable by a partial search. The obstruction is therefore not a failure to compute each individual unique equilibrium. It is the inability to recognise uniformly that no competing equilibrium exists, or to supply certain total convergence guarantees across both branches.

The physical model uses one distinguished Ising spin and a large finite auxiliary alphabet. It is not a two-state spin model. With dimension, finite supports, alphabet and a finite allowed set of energy values all fixed, there are only finitely many tables, so an unbounded halting reduction needs an input channel. Here the alphabet supplies that channel. Reducing its size, replacing the threshold multiplicity encoding by a smaller positive representation, or obtaining nearest-neighbour pair interactions without an implicit hard constraint remains open.

The dimension also has a clear role. A translation-invariant one-dimensional finite-table finite-energy interaction has a primitive transfer representation after finite blocking and hence a unique Gibbs state. Such a class cannot provide the coexistence branch used here. Uniform boundedness or translation invariance matters in this observation; merely requiring each term of a spatially unbounded inhomogeneous one-dimensional interaction to be finite is insufficient.

The soft construction simplifies the computational grammar and eliminates proof pins, but retains exact character cancellation. A genuinely different exact-integer finite-temperature

reduction remains desirable. Merely giving a finite program for a computable real coupling changes the representation: a small positive Ising field can then encode halting through its name, but an algorithm outputting its exact numerator and denominator would already decide the zero-versus-positive question. Such an encoding does not replace the finite-table theorem.

Several probabilistic extensions are unresolved. We do not prove analyticity of the target pressure, exponential convergence for arbitrary fixed microscopic auxiliary boundaries, strong spatial mixing, a spectral gap for Glauber dynamics, or robustness to all sufficiently small physical perturbations of a fixed model. The finite-object exchange controls equilibrium disagreements, not time-dependent relaxation. The reference Ising theorems giving analyticity or stronger mixing cannot simply be transferred to a target lacking their ferromagnetic structure. To settle these questions one would need, respectively, controlled growth of cumulant constants, a treatment of fixed auxiliary boundary layers, a dynamical coupling, or a quantitative norm bound on induced spin interactions under microscopic perturbation.

The central finite-certificate mechanism is more general than its implementation: exact positive realisation produces a canonical finite-object representation; one witnessed object supplies a uniform bias; an ordered reference controls one kind of defect; an independent sparse covering controls the remaining objects; finite exchange removes all competing canonical Gibbs states. The construction verifies this chain for exact finite-temperature tables and all physical Gibbs states. The constituent Ising, comparison, exchange and thermodynamic principles retain their established provenance.

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Appendix A. An explicit compiler for finite computation rectangles

This appendix gives two effective finite Wang-tile compilers. The first produces only accepting finite components and marks their top-left corners. The second permits every finite initial history and marks every occurrence of a halting head. The machine simulation and rectangle geometry are the same in both cases. These are statements about *finite components*; no exclusion of infinite matching configurations is asserted.

Proposition A.1 (Finite-component compiler)**.** *Given a deterministic Turing machine and a finite input, one can compute a finite set of labelled Wang tiles, a finite edge-colour alphabet with a distinguished colour* 0*, and a set of marked tile types, with the following properties. Adjoin a vacuum letter having colour* 0 *on all four edges. In any matching configuration on* $\mathbb{Z}^2$*, connect two nonvacuum tiles when their common edge has nonzero colour. Every finite connected component is a solid rectangle of width and height at least two, whose rows are successive states of the prescribed initial computation on a finite tape interval, with an absorbing rejection on overflow.*

(i) *In the* accepting version*, the top row contains a halting head, and the component has precisely one marked tile, its top-left corner. A finite component exists if and only if the given machine halts.*

(ii) *In the* history version*, no terminal acceptance condition is imposed. A tile is marked precisely when its centre letter carries a halting head. A finite* marked *component exists if and only if the given machine halts. Unmarked components are allowed.*

The output is obtained by enumerating finite sets and evaluating finite transition tables. In particular the compiler does not test whether a halting computation or a marked rectangle exists.

A.1. **A folded, bounded-width deterministic simulation.** Let Γ be the finite tape alphabet, with blank symbol $\square$. Normalize the machine so that it starts in state q_0 at tape position 0, the input is written at positions $0, 1, \ldots$, and all other positions are blank. Its tape is indexed by $\mathbb{Z}$. Replace all genuine halting outcomes by one absorbing state HALT. Complete the transition table on all other state–symbol pairs, using HALT for a missing transition if that is the original machine's halting convention. Add a *new* absorbing state REJECT, reserved for finite-interval overflow. Thus an original halting outcome is never confused with the overflow state. Allow moves in $\{-1, 0, +1\}$; an unused stay-put move causes no change to the construction.

Fold the two-sided tape onto positions $j \geq 0$. Position j stores a pair $(A_j, B_j) \in \Gamma^2$, representing original positions j and $-j-1$, respectively. A head has a state and a track $\tau \in \{A, B\}$. Each cell also has an immutable origin bit o, which is 1 at $j = 0$ and 0 elsewhere. An original transition writes only the currently selected component of the pair. Its physical motion and new track are as follows:

track	original move	physical move	new track
A	$+1$	$+1$	A
A	-1, $o = 0$	-1	A
A	-1, $o = 1$	0	B
B	-1	$+1$	B
B	$+1$, $o = 0$	-1	B
B	$+1$, $o = 1$	0	A
A or B	0	0	unchanged

The new machine state is the state prescribed by the original transition. For HALT and REJECT, the update preserves the head, track and tape symbols. One original step is one folded step, including a track change at the origin.

On an interval $\{0, \ldots, W-1\}$, a proposed physical move through a wall is instead replaced by a REJECT head at its source, after the prescribed write. Overflow takes precedence even if the original transition proposed the new state HALT. This may reject a narrow certificate, but never accepts an invalid one. Every finite genuine computation fits a sufficiently wide interval.

The finite cell alphabet is

$$\Sigma = \big(\Gamma^2 \times \{0,1\} \times (\{\text{no head}\} \cup (Q \times \{A, B\}))\big) \ \cup \ \{\bot_0, \bot_1\}, \tag{A.1}$$

where Q includes the two absorbing states. The poison letter $\bot_o$ retains origin bit o. Let $\#$ be a wall symbol not in Σ.

A.2. **A completely specified radius-one update.** Define a deterministic finite map

$$F : (\Sigma \cup \{\#\}) \times \Sigma \times (\Sigma \cup \{\#\}) \longrightarrow \Sigma. \tag{A.2}$$

Only triples whose centre lies in Σ are used. For a nonpoison head letter, its own state, selected tape symbol, track and origin bit determine its prescribed write, proposed new state and track, and physical displacement. These data are computable from that letter alone.

For a triple (p, x, q), apply the following rules in the displayed order.

(i) If any nonwall entry is poison, output the poison letter with the centre's origin bit.

(ii) Form the centre's new tape pair: perform its prescribed write if it has a head, and otherwise retain its old pair. Copy its origin bit.

(iii) An *incoming proposal* comes from p if p has a head whose physical displacement is $+1$, and from q if q has a head whose physical displacement is -1. Each proposal carries that head's prescribed new state and track. If there are two incoming proposals, or if an incoming proposal meets a centre that already carries a head, output poison with the centre's origin bit.

(iv) If there is exactly one incoming proposal, install its state and track on the centre's new tape pair.

(v) Otherwise, if the centre has no head, retain a headless centre. If it has a head with displacement zero, retain its prescribed new head. If it moves toward an actual neighbouring cell, leave its source headless. If it moves toward a wall #, retain an absorbing REJECT head at the source. Use the source track in this last case.

These alternatives define every triple in (A.2). For configurations reached from the specified one-head initial row, the poison cases never occur. A direct check shows that a head either stays, transfers to one neighbouring cell, changes track at the origin, or becomes the single absorbing overflow head. Therefore there remains exactly one head and the parallel update is precisely the folded finite-interval simulation. Soundness below uses this invariant, rather than any intended interpretation of malformed histories.

A.3. **The row automata.** Let $P = P_0 \cdots P_{\ell-1}$ be the folded input prefix, where $\ell \geq 1$ is large enough to include the entire input and the head at cell zero. At cell zero the origin bit is one and the initial head is (q_0, A); all other prefix cells are headless and have origin bit zero. Unused tape components are blank. Write z for the blank, headless, origin-bit-zero letter. The initial-row language is exactly

$$Pz^*. \tag{A.3}$$

Its deterministic finite automaton has states $0, 1, \ldots, \ell$ and a rejecting sink. At $i < \ell$, the only nonrejecting transition reads P_i and goes to $i+1$. At ℓ, the only nonrejecting transition reads z and stays at ℓ. The start state is 0 and the sole accepting state is ℓ. All other transitions go to the sink. In particular a proper prefix of P is not accepted.

For the accepting version, the top-row automaton has states u, a, d, meaning unseen, seen and dead. A headless nonpoison letter leaves u or a unchanged. A HALT head changes u to a, and changes a to d. Every other head and every poison letter goes to d; d remains d on every letter. The start state is u and the sole accepting state is a. Its language consists of words with exactly one halting head and otherwise ordinary headless letters.

For the history version, the top automaton has one accepting state with a loop on every letter of Σ. In both versions the interior-row automaton is this one-state automaton. The geometry below requires the origin bit of each centre letter to be one exactly in the leftmost column, independently of the row automata.

A.4. **The complete finite tile generator.** Use horizontal statuses $\eta \in \{L, I, R\}$ and vertical statuses $\nu \in \{B, I, T\}$. The letters L, R are distinct, as are B, T. A candidate tile is a tuple

$$(\eta, \nu, p, x, q, k_-, k_+), \tag{A.4}$$

where $x \in \Sigma$, $p, q \in \Sigma \cup \{\#\}$, and k_-, k_+ are states of the automaton assigned to row status ν. Enumerate all such tuples and retain exactly those satisfying:

(i) $p = \#$ if and only if $\eta = L$, and $q = \#$ if and only if $\eta = R$;

(ii) the origin bit of x is one if and only if $\eta = L$;

(iii) k_+ is the automaton transition from k_- on x, and neither state is rejecting or dead;

(iv) if $\eta = L$, then k_- is the start state;

(v) if $\eta = R$, then k_+ is an accepting state.

Put $y = F(p, x, q)$. Give the retained tile the colours

$$c_W = \begin{cases} 0, & \eta = L, \\ (\mathrm{hor}, \nu, p, x, k_-), & \eta \neq L, \end{cases} \qquad c_E = \begin{cases} 0, & \eta = R, \\ (\mathrm{hor}, \nu, x, q, k_+), & \eta \neq R, \end{cases} \tag{A.5}$$

$$c_S = \begin{cases} 0, & \nu = B, \\ (\mathrm{ver}, \eta, x), & \nu \neq B, \end{cases} \qquad c_N = \begin{cases} 0, & \nu = T, \\ (\mathrm{ver}, \eta, y), & \nu \neq T. \end{cases}$$

The two tags hor and ver are distinct formal symbols. Every displayed tuple is a nonzero colour. The colour 0 is different from every tuple, including tuples containing a blank, a wall symbol or the numerical automaton state 0.

In the accepting version mark a tile exactly when $(\eta, \nu) = (L, T)$. In the history version mark it exactly when x has a HALT head. The tile set consists of these retained labelled tuples. Distinct retained tuples can be kept as distinct labelled types even if their four colours happen to coincide. All subsequent tile counts and activities must count those labels consistently. No deduplication or minimization is needed for the theorem.

A.5. **Geometry, soundness and completeness.**

Proof of Proposition A.1. Take a finite nonzero-edge component and a tile at its lowest vertical coordinate. Its south edge must be zero, hence its vertical status is B. Following nonzero horizontal edges cannot continue indefinitely. It therefore yields an entire row starting with L, ending with R, and having only I statuses between them. Matching horizontal colours force one common vertical status and a matching automaton run along the row. A gap or a premature endpoint would put a zero colour against a nonzero colour. Because L and R are distinct, the row has width at least two.

Every tile of this bottom row has a nonzero north edge. Vertical matching produces the row above, carrying the same horizontal statuses and therefore exactly the same horizontal extent. Its common vertical status is I or T, since its south edges are nonzero. Repeat this argument. Finiteness forces a top row of status T. The distinction between B and T gives height at least two. The resulting rectangle has colour zero on its entire exterior boundary, so no further tile can connect to it. In particular no hole, branch, ragged row or extra piece of the component is possible. Other components can touch it only across zero-coloured edges.

Let $x_0 \cdots x_{W-1}$ be the centre letters in a row. Horizontal matching in (A.5) gives $p_i = x_{i-1}$ and $q_i = x_{i+1}$ at the internal positions, with $p_0 = q_{W-1} = \#$ at the two walls. The automaton states match as well. The bottom row is therefore exactly (A.3). Vertical matching forces each next centre letter to be $F(x_{i-1}, x_i, x_{i+1})$, with wall symbols at the ends. The origin-bit condition is correct at every height. The initial one-head invariant from Section A.2 now identifies every row with the genuine bounded-width computation. Poison cannot appear.

In the accepting version the top automaton requires a halting head. An overflow head remains REJECT forever, so acceptance certifies a genuine halt before any wall failure. Exactly one tile has statuses (L, T), giving the unique marked anchor. In the history version a mark is itself a genuine halting head in one of these rows. There is no requirement that such a head be unique among the rows: the absorbing halt may occur in several successive rows. Unmarked finite histories, including histories that eventually overflow, are permitted. Thus the indicated accepting or marked component always implies genuine halting.

Conversely suppose the original computation halts after finitely many steps. Choose $W \geq \max(2, \ell)$ large enough to contain all folded positions used up to and including that transition. The finite-interval simulation then reaches HALT without overflow. Stack its initial row and all successive images through a halting row, repeating the absorbing halt if needed to make the height at least two. Assign statuses B to the first row, T to the last, I between them, and L, I, R to the columns. The stored triples and the row automata give exactly the retained tiles above. They match vertically and horizontally, and their exterior colours are zero. This is a finite accepting component in the first version and a finite marked history component in the second. Every step of the construction is finite and effective, completing the proof. □

Remark A.2 (The distinction between the two versions). On a nonhalting input the accepting version has no finite nonvacuum component. The history version generally has many finite components, but none is marked. Consequently its positive object partition function is spin

independent in the nonhalting case, rather than necessarily equal to one. The accepting version supports a hard single-anchor spin eligibility. The history version uses a positive marked-site penalty in the main construction, which allows multiple marks without allowing a minus spin at a different mark to forbid the entire witness. This appendix changes neither the stated physical integer energies nor the finite-energy requirement: tile matching is used only in the proof representation.

Remark A.3 (Infinite sectors are not excluded). The all-interior blank tile is a simple example of a tile that can match itself on the whole plane. Such an infinite component is compatible with the grammar even on a halting input. The proposition therefore does not assert uniqueness of the hard-tile specification under arbitrary tile boundaries. In the physical model, arbitrary Gibbs states are related to the canonical finite-object law by a separate positive auxiliary-variable mixing argument; the finite compiler alone does not justify that passage.

Appendix B. Counting and quantitative covering estimates

B.1. **Rooted counts and collars.** A connected n-site support containing x has a deterministic spanning tree. A fixed rooted traversal of that tree is a nearest-neighbour walk of length $2(n-1)$ and determines its visited support. There are at most $4^{2(n-1)} \le 16^n$ such walks. Multiplication by t^n bounds labelled objects. Their diameter is at most n, so

$$\sup_x \sum_{\gamma:\, x\in S_\gamma} |S_\gamma| w_\gamma 2^{\operatorname{diam} S_\gamma} \le \sum_{n\ge 1} n(32u)^n.$$

For a connected n-edge dual graph, a deterministic traversal of its doubled edges gives at most 16^n graphs through a specified dual vertex. If its filled hull contains a fixed primal site, one can choose a dual vertex in a square of at most $(2n+3)^2$ possible locations. The hull has diameter at most n and cardinality at most $(n+1)^2$. These intentionally loose bounds include four-valent crossings and nested hulls.

For either object supports or hulls, one-step inflation costs at most five in rooted count, five in cardinality, and four in the weight 2^{diam}. Two replicas therefore cost at most $2\cdot 5\cdot 5\cdot 4 = 200$. This proves the two series in (4.7). The first factor five comes from choosing a site in the nearest-neighbour neighbourhood of the query lying in the uninflated support; it is distinct from the cardinality factor five.

B.2. **Exact numerical bounds.** Put $a = 32/10000 = 2/625$. Then

$$2\sum_{n\ge 1} na^n = \frac{2500}{388129} < 0.006442, \qquad 200\sum_{n\ge 1} na^n = \frac{250000}{388129}.$$

The partial exponential series gives $e > 163/60 > 19/7$, and $(19/7)^8 > 2560$. Thus, for $K \ge 8$, $32e^{-K} < 1/80$. If $P_n = (n+1)^2(2n+3)^2$, then for $n \ge 4$,

$$\frac{P_{n+1}}{P_n} \le (6/5)^2(13/11)^2 < 9/4, \qquad P_4 = 3025.$$

Consequently

$$200\sum_{n\ge 4} P_n(32e^{-K})^n < \frac{200\cdot 3025\cdot 80^{-4}}{1-9/320} = \frac{605}{39808},$$

and

$$B < b_0 := \frac{250000}{388129} + \frac{605}{39808} = \frac{10186818045}{15450639232} < \frac{2}{3}. \tag{B.1}$$

These bounds use no temperature-dependent choice of the physical table. They prove the entire finite half-line $\beta \ge 2/5$ for the soft table. There is no assertion at $\beta = \infty$, where positivity is lost, or that $2/5$ is optimal.

The minority probability in the Ising plus phase does not require the hull-cardinality factor. By the contour covering,

$$\delta_I \le \sum_{n\ge 4} (2n+3)^2 (16e^{-K})^n.$$

The polynomial ratio is below 2 for $n \ge 4$. At $K \ge 8$,

$$\delta_I < \frac{121 \cdot 160^{-4}}{1 - 2/160} = \frac{121}{647168000} < 10^{-6}. \tag{B.2}$$

At $K = 7$, the weaker $e > 2$ gives $\delta_I < 121/3072 < 1/2$, establishing coexistence below the reference values used in the main theorems. At the hard table's $K = 20$, $16e^{-20} < 2^{-16}$ gives

$$\delta_I < \frac{121 \cdot 2^{-64}}{1 - 2^{-15}} < 10^{-16}.$$

In particular the hard table's equal-mixture spin covariance is at least $1 - 4\delta_I > 0.99$ at every separation, even without association: a union bound in the plus phase gives $\mathbb{E} s_0 s_x \ge 1 - 4\delta_I$, and spin reversal leaves that two-point function unchanged.

B.3. **Weighted chains and one-arm bounds.** For clarity, let independent Bernoulli species have parameters z_ξ and finite supports T_ξ, and suppose

$$b = \sup_x \sum_{\xi:\, x \in T_\xi} |T_\xi| z_\xi 2^{\operatorname{diam} T_\xi} < 1.$$

For a distinct-species intersection chain from x to y, the sum of support diameters is at least $\operatorname{dist}(x, y)$. Independence bounds its probability by the product of parameters. To sum chains, choose an initial support containing x; choose an exit site in it; choose the next support containing that site; and continue. After multiplying by $2^{\sum \operatorname{diam}}$, every step is bounded by b. A final factor $|T_\xi| \ge 1$ can be inserted harmlessly. Thus chains of length k contribute at most $b^k 2^{-\operatorname{dist}(x,y)}$.

For an arm to distance R, the same argument stops at its first reach of that distance. It yields $\sum_{k\ge 1} b^k 2^{-R}$ directly, without summing over all possible endpoints. Independence was applied only to distinct species; the geometric numerical sum is enlarged by allowing repetitions. Local finiteness follows from $\sum_{\xi: x \in T_\xi} z_\xi < \infty$. The one-arm estimate excludes infinite spatial components; local finiteness then excludes infinitely many species in a finite component. These statements justify both the finite gas exchange and the finite replica swaps.

Appendix C. Exact interfaces with established theorems

This appendix records the mathematical hypotheses of the imported results and where they hold. The direct arguments in the text are retained so that no unproved transfer from the reference Ising model to the target is implicit.

C.1. **Conditional Dobrushin comparison.** Rebeschini–van Handel [21, Corollary 2.6, arXiv version 1] gives the classical single-site comparison bound. With influence matrix C, local update discrepancies b_j, and $D = \sum_{n\ge 0} C^n$, the difference of expectations is bounded by $\sum_{i,j} \operatorname{osc}_i(f) D_{ij} b_j$, subject to quasilocal reference kernels and the stated vanishing-C^n condition. Finite discrete single-site spaces satisfy its metric and topological hypotheses. This is a modern precise formulation of the classical Dobrushin theorem [6].

Here the index set is the countable edge set of $\mathbb{Z}^2$, each site space is a finite group, the single-edge heat-bath kernels are local and positive, and (3.3) gives $\|C\|_\infty \le \alpha < 1$. Thus row sums of C^n are at most α^n and the vanishing condition holds. Equal specifications have $b = 0$, giving uniqueness. Remote boundary or spin modifications create discrepancies only where the local factors change, so the resolvent sum starts at the corresponding path distance. Conditioning on a finite set, or local positive tilting followed by approximation,

gives the conditional covariance consequences used in (4.10). Multiplicity restoration is a local independent kernel. None of this compares the low-temperature spins by Dobrushin's criterion.

C.2. **Positive-field Ising and zero-field classification.** Lebowitz–Martin-Löf [18, Theorem 1] treats translation-invariant summable nonnegative Ising pair interactions with a uniform field (and allows a periodic background). It identifies uniqueness with differentiability in the uniform field. Its explicit consequence on p. 278, from Lee–Yang [19], is uniqueness at nonzero homogeneous field when the background is zero. Their boundary comparison, in particular Lemma 2, sandwiches arbitrary finite-boundary correlations between plus and minus limits. It supplies the all-DLR formulation: condition any DLR law on a box and let the equal plus/minus limits exhaust it.

Our reference has nearest-neighbour $J = K/2 > 0$, absolutely summable nonnegative pair couplings, zero background and uniform $\delta > 0$. Every hypothesis therefore holds. Reference monotonicity and the limit as $\delta \downarrow 0$ identify the plus law used in (4.4). Holley's finite-lattice criterion [13] is a standard comparison route, but we do not leave a target/reference lattice inequality unchecked: the ordered heat-bath proof in section 4.3 proves precisely the needed comparison without target FKG.

Coquille–Velenik [4, Corollary 1.1] states that for the nearest-neighbour zero-field ferromagnetic Ising model on $\mathbb{Z}^2$ at $J > J_c$, the set of *all* DLR measures is exactly the segment between the plus and minus states. No translation-invariance hypothesis on the input state is required. Our nonhalting projection is that exact specification, and appendix B verifies $J \geq 4 > J_c$. The affine inverse lift in section 3.3 transfers the entire segment, extremality and translation invariance. This classification is an application of Aizenman–Higuchi, not a new general Ising theorem.

C.3. **Independent disagreement and the object gas.** Van den Berg [22, Theorem 1, p. 162] assumes a countable connected locally finite graph, finite or countable site spaces, and two Markov fields with the same specification. If their independent product has no infinite disagreement path, the laws coincide. Its proof on p. 163 exchanges finite disagreement clusters. Our graph is $\mathbb{Z}^2$, the spin-and-tile site space is finite, and lemma 3.2 verifies the common nearest-neighbour specification for all canonical physical lifts. The independent covering estimate then verifies absence of infinite disagreements. The direct proof in lemma 4.1 is this established exchange, with its covariance consequence. It is applied pairwise within the canonical class, not to unrelated infinite-component tile states.

Van den Berg–Maes [23, Theorem 1 and Corollaries 1–2] provide finite-volume disagreement coupling and percolation criteria using worst-conditional one-site discrepancies. Their small-percolation criterion is not automatically met here. At $\beta = 1$, vacuum neighbouring tiles and all-plus versus all-minus neighbouring spins already give spin-conditional variation $\tanh(40)$. Our upper families instead describe rare typical defects after a phase comparison.

For an alternative canonical-gas construction, Ferrari–Fernández–Garcia [8, Theorem 1(ii)] constructs the stationary birth/death process from finite free-process ancestor clans. Their discrete branching criterion in equations (4.3)–(4.4) is

$$\sup_{\gamma} \frac{1}{q(\gamma)} \sum_{\eta} q(\eta)\lambda_\eta I(\gamma,\eta) < 1, \qquad \inf_{\gamma} q(\gamma) \geq 1.$$

Take countably many finite labelled objects, self-including overlap incompatibility, death rate one, birth rates $a_\eta(s) \leq w_\eta$, and $q(\gamma) = |S_\gamma|$. The left side is at most

$$\sup_{x} \sum_{\eta:\, x \in S_\eta} |S_\eta| w_\eta \leq \sum_{n \geq 1} n(16u)^n < 0.001606.$$

Rooted free birth intensities are finite. Common Poisson cylinders therefore construct finite ancestor clans for any finite query. Detailed balance identifies finite restrictions with the

product-activity gas. Common ancestor constructions remove finite cutoffs; the diameter-weighted bound $\sum_n n(32u)^n < 0.003221$ gives exponential locality by generation paths. These last quantitative estimates are the applicability calculation, not extra statements quoted from that theorem. The incompatibility graph has infinite degree, so a locally finite graph theorem should not be applied directly on its species vertices.

The Section 2 Theorem of Kotecký–Preiss [15] is another optional route. With size function $a(\gamma) = |S_\gamma|$, its incompatibility summability criterion holds even for activities inflated by $e^{|S_\gamma|}$, since

$$\sum_{\eta\sim\gamma} w_\eta e^{2|S_\eta|} \le |S_\gamma| \sum_{n\ge1} (16ue^2)^n < |S_\gamma|.$$

Here $u < 10^{-4}$ and $e < 3$ suffice; deleting or reducing activities preserves the bound. We use neither the resulting logarithmic expansion nor an induced-potential reconstruction. The static finite exchange is sufficient.

C.4. **Reference boundary estimate.** Ott [20, Theorem 6.1, arXiv:1905.12928v2] fixes nearest-neighbour ferromagnetic Ising on $\mathbb{Z}^d$ with exponent coupling nonnegative and uniform $h > 0$. It bounds the plus/minus centre magnetization difference in $[-N, N]^d$ by $Ce^{-\nu N}$, where $C < \infty$ and $\nu > 0$ depend on the reference parameters. We use $d = 2$, coupling $K/2$, field $h_* > 0$, strictly within these hypotheses. Inner-box monotonicity gives (5.2). The query-buffer and swap proof of (5.3) is supplied here; the target is not itself an Ising model. Ott's analyticity theorems are not imported for the target.

C.5. **Variational pressure and smoothness.** Barbieri–Gómez–Marcus–Taati [1, Theorem 1.1(a),(b), Proposition 3.2; arXiv version 2] supplies the Gibbs/equilibrium and pressure interfaces. The Gibbs-to-equilibrium direction assumes a finite-alphabet $\mathbb{Z}^d$ subshift with D-mixing and an absolutely summable interaction; the reverse direction assumes finite type. Proposition 3.2 uses a standard Borel environment, relative topological Markov property and D-mixing along a Følner sequence. For us the environment is a singleton, the physical configuration space is the finite full shift, all patterns paste freely, boxes are Følner, and the interaction is finite range. Every hypothesis is immediate on this *physical* shift; no such claim is needed for constrained proof tiles. It follows that the free partition-function limit is the invariant variational pressure and that its maximisers are precisely invariant physical Gibbs states. Equation (6.2) is then proved directly.

Lebowitz [17, Section I, (1.12)–(1.13); Section II, (2.4a,b), pp. 317–318] is a predecessor for fixed-order Ursell decay and smoothness from uniformly summable derivatives; this general discussion is unnumbered. Lebowitz–Penrose [16, Section VII, (90)–(93)] supplies a further classical thermodynamic precedent. The target's finite-support inhomogeneous-tilt argument is proved in section 6.3, rather than assuming ferromagnetic hypotheses it does not have.

Gayral–Sablik–Taati [9, Proposition 14] proves computability of pressure for computable potentials on finite-alphabet full shifts; the present table meets those hypotheses. Equation (6.1) is a direct elementary certificate in this special class. Their Proposition 15 concerns effective invariant equilibrium sets, rather than a chosen microscopic boundary sequence. Fawzi–Fawzi–Scalet [7, Theorems 2.1–2.4, published numbering] gives certified equilibrium-observable intervals, invariant refinements, a strict all-states sign promise algorithm, and quantitative commuting high-temperature/one-dimensional bounds. Our diagonal interaction fits the commuting setting, but its nonhalting magnetization signs violate that strict sign promise. Target clustering alone does not meet their high-temperature criterion. No quantitative result from these theorems is used in our finite-volume proof.

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